Spatial and connectivity

SWAT+ Documentation

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Purpose

The connect files describe how water and the material it moves travel from one spatial object to the next. Every spatial object (HRU, routing unit, aquifer, channel, reservoir, recall, export coefficient, delivery ratio, outlet, gwflow cell) has one row in the connect file for its object type. The row gives the object its number and name, its area, location and elevation, the record it uses in its property file, its weather station, and the list of objects that receive its outflow. SWAT+ builds the routing network and the order in which objects are run from these rows.

All connect files have the same format. The file names are given on the `connect` line of `file.cio`, in this order:

Position on the `connect` line	Default name	Object	Count column in `object.cnt`	Property column header	Property file
1	`hru.con`	HRU	`hru`	`hru`	`hru-data.hru`
2	`hru-lte.con`	HRU-lte	`lhru`	`hlt`	`hru-lte.hru`
3	`rout_unit.con`	Routing unit	`rtu`	`rtu`	`rout_unit.rtu`
4	`gwflow.con`	gwflow cell	`mfl`		see gwflow
5	`aquifer.con`	Aquifer	`aqu`	`aqu`	`aquifer.aqu`
6	`aquifer2d.con`	2-D aquifer	`aqu2d`		not read in 62.0.1
7	`channel.con`	Channel (legacy)	`cha`		`channel.cha`
8	`reservoir.con`	Reservoir	`res`	`res`	`reservoir.res`
9	`recall.con`	Recall (point source or inlet)	`rec`	`rec`	`recall.rec`
10	`exco.con`	Export coefficient	`exco`	`exc`	`exco.exc`
11	`delratio.con`	Delivery ratio	`dlr`	`dlr`	`delratio.del`
12	`outlet.con`	Outlet	`out`	`out`	none
13	`chandeg.con`	SWAT-DEG channel	`lcha`	`lcha`	`channel-lte.cha`

A file is read only when the matching count in object.cnt is above zero. When a count is above zero the file has to exist, because nothing checks for it: a missing file leaves those objects without connection data. The file named in position 6 is never opened in 62.0.1.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one object. SWAT+ reads as many rows as the count for that object type in object.cnt. It does not match columns by header name, so every row must give the values below in this order.

Each row has 13 leading values followed by out_tot groups of four values that describe the outflow connections, all on the same line.

#	Column	Units	Description
1	id	-	Object number within its type. Number the rows 1, 2, 3 and so on in file order: the obj_id column of other connect files refers to an object by its row position, while SWAT+ uses id as the object number during the run.
2	name	-	Object name, up to 16 characters. It appears in the output files and is not used in any calculation.
3	gis_id	-	Object number in the GIS interface. Not used in calculations. The SWAT+ Editor renumbers objects consecutively when it writes the files, so gis_id and id often differ; use id when matching a channel to a stream gauge.
4	area	ha	Area of the object. See Area, location and elevation.
5	lat	decimal degrees	Latitude.
6	lon	decimal degrees	Longitude. Not used in calculations.
7	elev	m	Elevation.
8	property column (hru, hlt, rtu, aqu, lcha, res, rec, exc, dlr, out)	-	Row number in the property file listed in the table above. Called obj in earlier documentation. In recall.con this column is not used: a recall object uses the recall.rec record whose id equals the recall object's id.
9	wst	-	Weather station name from weather-sta.cli. See Weather station lookup.
10	cst	-	Constituent pointer. Read but not used.
11	ovfl	-	Overbank flooding pointer. Read but has no effect in 62.0.1.
12	rule	-	Name of a flow-control decision table in flo_con.dtl, or null. See Flow-control rule.
13	out_tot	-	Number of outflow connections that follow on the line. May be 0.

For each of the out_tot connections, four values follow:

Offset	Column	Units	Description
+1	obj_typ	-	Type code of the receiving object. See Receiving object codes.
+2	obj_id	-	Row number of the receiving object in its connect file.
+3	hyd_typ	-	Which outflow hydrograph of this object is sent. See Hydrograph codes.
+4	frac	fraction	Fraction of that hydrograph sent to the receiving object.

When out_tot is 0 the row ends after column 13 and the object sends no outflow through this file. HRUs that belong to a routing unit are usually written this way, because the routing unit collects their outflow (see rout_unit.def and rout_unit.ele).

Area, location and elevation

The `area` value is used in calculations for some object types only:

Object	Use of `area`
HRU	Area of the HRU. Converts depths (mm) to volumes (m^3) and loads per hectare to totals.
Routing unit	Area of the routing unit. Converts inflow volumes to depths when water is routed across it.
Aquifer	Area of the aquifer. Converts recharge, storage and seepage between mm and m^3.
HRU-lte	Area of the HRU-lte. Converts runoff and sediment to totals.
SWAT-DEG channel	Used with `aqu_cha.lin` to order the channels linked to an aquifer by drainage area, and to express aquifer inflow as a depth in the channel output. See link files.
Other objects	Not used in calculations.

At the start of the run SWAT+ also adds up the drainage area of every object from the areas of the HRUs, routing units and recall objects upstream of it, weighted by `frac`, and writes the computed and the input areas to `area_calc.out`. Comparing the two columns in that file is a quick check of the routing.

`elev` is used for every object to adjust precipitation and temperature for the difference in elevation between the object and its weather data. With the lapse rates `plaps` (mm per km) and `tlaps` (°C per km) from `parameters.bsn`,

$$\Delta P = \text{plaps}\,\frac{z_{obj} - z_{pcp}}{1000}, \qquad \Delta T = \text{tlaps}\,\frac{z_{tmp} - z_{obj}}{1000}$$

where z_{obj} is `elev` and z_{pcp} and z_{tmp} are the elevations of the precipitation and temperature gauges, or of the weather generator station when the variable is simulated. For HRUs `elev` is also the elevation used to compute atmospheric pressure for potential evapotranspiration, and it can be changed in calibration with the parameter `elev`.

`lat` matters only for HRUs: an HRU with a negative latitude is treated as being in the Southern Hemisphere, and its yearly reset of base heat units and of `skip` operations happens on day 181 instead of 31 December.

Weather station lookup

The `wst` names are looked up in the list of stations read from `weather-sta.cli` with a binary search. The search assumes the stations are listed in ascending order of name (character-code order). If the stations are not in that order, or the name is not in the file, the object uses the first station in `weather-sta.cli` and no message is written.

Receiving object codes

`obj_typ` accepts these codes:

Code	Receiving object	Notes
`hru`	HRU	Inflow is routed across the HRU. See Inflow to HRUs and routing units.
`hlt`	HRU-lte	
`ru`	Routing unit	The routing unit and each HRU in it are registered as receivers.
`aqu`	Aquifer	
`cha`	Channel in `channel.con`	62.0.1 does not run these legacy channels, so flow sent to them does not continue downstream. Use `sdc` for SWAT-DEG channels.
`res`	Reservoir	

Code	Receiving object	Notes
`exc`	Export coefficient	Export coefficient objects are not run as part of the routing, so inflow to them has no effect.
`dr`	Delivery ratio	The object multiplies its inflow by the ratios in `delratio.del`.
`out`	Outlet	
`sdc`	SWAT-DEG channel in `chandeg.con`	

A recall object cannot receive flow. The routing code also has a branch for gwflow cells as receivers, but its code is longer than the three characters `obj_typ` can store, so no value in a connect file can select it. Any other code is not recognised and leaves the connection without a receiving object, which is likely to stop the run.

Hydrograph codes

`hyd_typ` selects one of the outflow hydrographs of the sending object:

Code	HRU, HRU-lte, routing unit	Aquifer	Channel, reservoir, recall, outlet, export coefficient, delivery ratio
`tot`	Surface runoff + lateral flow + tile flow	Return flow (groundwater discharge)	Total outflow
`rhg`	Percolation out of the bottom of the soil profile (recharge)	Deep seepage	Not filled, sends nothing
`sur`	Surface runoff	not used	not used
`lat`	Lateral soil flow	not used	not used
`til`	Tile flow	not used	not used

Any other code is not recognised. The usual connections are `tot` from an HRU or routing unit to a channel together with `rhg` to its aquifer, and `tot` from an aquifer to a channel.

`frac` is not checked. The fractions of one hydrograph type from one object should add up to 1, otherwise water is created or lost. Fractions of different types are independent, which is why a routing unit can send 100% of `tot` to a channel and 100% of `rhg` to an aquifer. Sending both `tot` and one of its parts (`sur`, `lat`, `til`) counts that part twice.

Inflow to HRUs and routing units

An HRU or routing unit that receives flow keeps surface, lateral and tile inflow separate, so it can route each through the right part of the landscape:

- From another HRU, HRU-lte or routing unit with `tot`, the sender's surface runoff is added to surface runon, its lateral flow to lateral inflow and its tile flow to tile inflow.
- From another HRU, HRU-lte or routing unit with `sur`, `lat` or `til`, that hydrograph goes to surface runon, lateral inflow or tile inflow. An `rhg` connection to an HRU or routing unit is accepted, but the water is not added anywhere.
- From any other object (channel, aquifer, reservoir, recall and so on) the total outflow is added to surface runon, whatever `hyd_typ` says.

For all other receiving objects the selected hydrograph, multiplied by `frac`, is added to the object's inflow.

Flow-control rule

A `rule` name other than `null` is matched against the decision tables in `flo_con.dtl` when the files are read. In 62.0.1 the match is stored but no routine uses it, so `rule` does not change the outflow fractions during the run.

Export coefficient objects

In `exco.con` the `exc` column is the row number in `exco.exc`. The `om` name in that row selects a row of `exco_om.exc`, and that row becomes the object's outflow hydrograph. It is set once when the files are read and stays constant for the whole run. Send it on with `hyd_typ tot`.

Groundwater flow (gwflow) models

When the gwflow module is active (`gwflow` = 1 in `codes.bsn`) and `object.cnt` lists gwflow cells, any row that has at least one connection with `obj_typ aqu` has its connection count reduced by one after reading, because gwflow replaces the aquifer objects. The connection that is dropped is the last one on the line, which is the aquifer connection only when that connection is written last. Put aquifer connections at the end of the line in gwflow models.

Routing loops

SWAT+ orders the objects so that every object runs after all objects that send it flow. If the connections contain a loop, the ordering cannot finish: after 5000 passes SWAT+ writes the objects that could not be ordered, with their inflow objects, to `looping.con`, prints an error and stops.

Example

`refdata/Osu_1hru/aquifer.con` in the SWAT+ repository:

```
aquifer.con: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id  name                gis_id          area           lat           lon          elev       aqu
       1  aqu001                   1      10.00000      35.64206     127.38513     322.26395         1
```

One aquifer of 10 ha that uses row 1 of `aquifer.aqu` and sends all of its return flow (`tot`) to SWAT-DEG channel 1.

`refdata/Osu_1hru/rout_unit.con`:

```
rout_unit.con: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id  name                gis_id          area           lat           lon          elev       rtu
       1  rtu001                   1       10.0000      35.52014     127.32787     113.51276         1
```

The routing unit sends its surface, lateral and tile flow (`tot`) to SWAT-DEG channel 1 and its recharge (`rhg`) to aquifer 1, each with fraction 1.0. The `rule` value `0` does not name a table in `flo_con.dtl`, so it has the same effect as `null`.

`refdata/Osu_1hru/hru.con`:

```
hru.con: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id  name                gis_id          area           lat           lon          elev       hru
       1  hru0001                  1           10       35.65311     127.36763     350.96629         1
```

The HRU has no connections in `hru.con` (`out_tot` = 0). Its outflow reaches the channel and the aquifer through the routing unit it belongs to.

Related files

- `object.cnt`: number of objects of each type, and so the number of rows read from each connect file.
- `file.cio`: names the connect files on its `connect` line.
- `hru-data.hru`, `aquifer.aqu`, `reservoir.res`, `recall.rec` and the other property files named in the first table.
- `aqu_cha.lin`: read right after `aquifer.con` to link an aquifer to more than one channel.
- `weather-sta.cli`: station names used in `wst`.

- Decision tables: `flo_con.dtl`, named in `rule`.
- `parameters.bsn`: lapse rates applied with `elev`.
- Recall, export coefficient and delivery ratio objects in the model theory.

Sources

- Reader: `hyd_read_connect.f90`, called once per object type from `hyd_connect.f90`, which also resolves receiving objects, hydrograph codes, drainage areas and the run order
- Type definitions: `type object_connectivity`, `type spatial_objects`, `type object_total_hydrographs` in `hydrograph_module.f90`
- File names: `type input_con` in `input_file_module.f90`, read in `readcio_read.f90`
- Inflow summing and hydrograph use: `command.f90`, `hru_hyds.f90`, `aqu_1d_control.f90`
- Area, elevation and latitude: `hrudb_init.f90`, `topohyd_init.f90`, `cli_lapse.f90`, `et_pot.f90`, `time_control.f90`
- Weather station lookup: `search.f90`, station list from `cli_staread.f90`
- Rule lookup: `dtbl_flocon_read.f90`
- Export coefficient hydrographs: `exco_read_om.f90`
- `aqu_cha.lin` reader: `aqu2d_read.f90`
- `area_calc.out` unit opened in `proc_bsn.f90`

Column	Fortran variable
`id`	`ob(i)%num`
`name`	`ob(i)%name`
`gis_id`	`ob(i)%gis_id`
`area`	`ob(i)%area_ha`
`lat`	`ob(i)%lat`
`lon`	`ob(i)%long`
`elev`	`ob(i)%elev`
property column	`ob(i)%props`
`wst`	`ob(i)%wst_c` (resolved to `ob(i)%wst`)
`cst`	`ob(i)%constit`
`ovfl`	`ob(i)%props2`
`rule`	`ob(i)%ruleset` (resolved to `ob(i)%flo_dtbl`)
`out_tot`	`ob(i)%src_tot`
`obj_typ`	`ob(i)%obtyp_out(k)`
`obj_id`	`ob(i)%obtypno_out(k)` (resolved to `ob(i)%obj_out(k)`)
`hyd_typ`	`ob(i)%htyp_out(k)` (resolved to `ob(i)%ihtyp_out(k)`)
`frac`	`ob(i)%frac_out(k)`

Purpose

`hru-data.hru` lists the HRU property sets of the project. Each row names one record in each of the parameter files an HRU needs: topography, hydrology, soil, land use and management, soil and plant initialization, surface storage (wetland), snow and field. An HRU picks its row through the `hru` column of `hru.con`, which gives the `id` of a row in this file. The file is named on the `hru` line of `file.cio` (default `hru-data.hru`).

The parameter values themselves are in the files the names point to. This file only links them together, so several HRUs can share one hydrology or topography record while having different soils or land uses.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one property set with ten values in the order below. The reader does not match columns by header name, so all ten values must be present and in this order.

#	Column	Looked up in	Description
1	id	-	Number of the property set. The hru column of hru.con refers to this value. The numbers do not have to be consecutive; the largest id sets the size of the table.
2	name	-	Name of the property set, up to 40 characters. Not used in calculations.
3	topo	topography.hyd	Topography record: slope, slope lengths and deposition coefficient.
4	hydro	hydrology.hyd	Hydrology record: lateral flow, canopy storage, evaporation and percolation parameters.
5	soil	soils.sol	Soil profile.
6	lu_mgt	landuse.lum	Land use and management record, which sets the plant community, management schedule, curve numbers, conservation practices and structures such as tile drains.
7	soil_plant_init	soil_plant.ini (see initial conditions)	Initial soil water, soil nutrients and constituents. The names of the soil test, pesticide, pathogen, heavy metal, salt and constituent records given in that row are matched at the same time.
8	surf_stor	wetland.wet	Surface storage (wetland or paddy) record, or null for an HRU without surface storage.
9	snow	snow.sno	Snow parameter set.
10	field	field.fld	Field dimensions, or null to use the built-in values.

A name must match the record name in the target file character for character, including case. Names in topography.hyd, hydrology.hyd and field.fld are stored with at most 16 characters and soil names with at most 20, so a longer name in this file can never match a record there.

Name lookup

Each name is looked up once, when the files are read. The table gives what happens when a name is not found:

Column	null or a name that is not found
topo, hydro, soil, lu_mgt, soil_plant_init, snow	The HRU uses the first record of the target file. No message is written.
surf_stor	null means no surface storage. Any other name that is not in wetland.wet is reported in diagnostics.out as <name> not found (wetland.wet) and the HRU has no surface storage.
field	The HRU uses the built-in field values (length 500 m, width 100 m, angle 30°). A name other than null that is not found is reported in diagnostics.out as <name> not found (field.fld).

The reader has checks that would write not found messages for topo, hydro, soil, lu_mgt, soil_plant_init and snow, but they test for a pointer of zero while the pointers start at one, so in 62.0.1 these messages are never written. Check these names by hand, or in the SWAT+ Editor, before a run.

For snow this means that null selects the first row of snow.sno, the same as an unmatched name. A project that has no snow.sno file at all has no first row to fall back on.

The surf_stor name is matched when the wetlands are initialized, which happens only when wetland.wet has at least one record.

Tile drainage

The lu_mgt record decides whether an HRU has tile drains. For a tile-drained HRU, SWAT+ replaces two values from its hydro record at the start of the run: cn3_swf is set to 0.95 and perco to 0.1. Calibration changes to

`cn3_swf` and `perco` are also skipped for these HRUs. See hydrology.hyd.

Example

`refdata/Ames_sub1/hru-data.hru` in the SWAT+ repository (first four rows):

`hru-data.hru:`

id	name	topo	hydro	soil	lu_mgt	soil_
1	hru0001	topohru0001	hyd0001	soil_01-h1	cosy_lum	
2	hru0002	topohru0002	hyd0002	soil_02	mntill_corn_lum	
3	hru0003	topohru0003	hyd0003	soil_03-h3	cosy_lum	
4	hru0004	topohru0004	hyd0004	soil_04	cosy_lum	

Each HRU has a separate topography record, hydrology record and soil. HRU 2 uses the land use `mntill_corn_lum`, the others `cosy_lum`. No HRU has surface storage, all use the snow set `snow001`, and `field` is `null`, so every HRU uses the built-in field dimensions.

Related files

- hru.con: one row per HRU; its `hru` column selects a row of this file by `id`. The HRU's area, elevation, latitude and weather station come from hru.con.
- object.cnt: number of HRUs.
- topography.hyd, [hydrology.hyd](#input-reference-hyd-hydrologyhyd) and field.fld, soils.sol, landuse.lum, initial conditions, wetland.wet, snow.sno: the files the names point to.
- file.cio: names this file on its `hru` line.
- Water allocation: an HRU can be named as a point of diversion in point_of_diver.wro. That link is made in the water allocation files; there is no column for it here.

Sources

- Reader: hru_read.f90, called from proc_hru.f90
- Type definitions: `type hru_databases`, `type hru_databases_char`, `type hydrologic_response_unit_db` in hru_module.f90
- Copy to each HRU through `hru.con`: hrudb_init.f90
- Use of the topography, hydrology, snow and field pointers, and the tile-drain override: topohyd_init.f90
- Surface storage lookup: wet_initial.f90, called from wet_all_initial.f90
- Name lengths of the target records: hydrology_data_module.f90, topography_data_module.f90, soil_data_module.f90, landuse_data_module.f90
- Points of diversion: water_allocation_read.f90
- File name: `in_hru%hru_data` in input_file_module.f90

Column	Fortran variable	Resolved pointer
`id`	array index `i` of `hru_db`	
`name`	`hru_db(i)%dbsc%name`	
`topo`	`hru_db(i)%dbsc%topo`	`hru_db(i)%dbs%topo` into `topo_db`
`hydro`	`hru_db(i)%dbsc%hyd`	`hru_db(i)%dbs%hyd` into `hyd_db`
`soil`	`hru_db(i)%dbsc%soil`	`hru_db(i)%dbs%soil` into `soildb`
`lu_mgt`	`hru_db(i)%dbsc%land_use_mgt`	`hru_db(i)%dbs%land_use_mgt` into `lum`
`soil_plant_init`	`hru_db(i)%dbsc%soil_plant_init`	`hru_db(i)%dbs%soil_plant_init` into `sol_plt_ini`
`surf_stor`	`hru_db(i)%dbsc%surf_stor`	`hru(j)%dbs%surf_stor` into `wet_dat` (set in `wet_initial`)
`snow`	`hru_db(i)%dbsc%snow`	`hru_db(i)%dbs%snow` into `snodb`
`field`	`hru_db(i)%dbsc%field`	`hru_db(i)%dbs%field` into `field_db`

Purpose

`hru-lte.hru` describes the HRU-lte objects, a simplified kind of HRU. An HRU-lte has one soil layer described by a texture class, a single plant whose growing season is started and ended by decision tables, a curve number for surface runoff, separate shallow and deep groundwater stores and a fixed USLE factor for erosion. It computes no nutrients, pesticides or bacteria. HRU-lte objects can be used alone or together with ordinary HRUs, and they send water and sediment to channels, routing units and other objects like any other spatial object.

The file is named in the second position of the `hru` line of file.cio (default `hru-lte.hru`). The objects themselves are defined in `hru-lte.con`, the second file on the `connect` line, and their number is the `lhru` count in object.cnt. The `hlt` column of `hru-lte.con` gives the `id` of the `hru-lte.hru` row each object uses. The area, latitude, elevation and weather station of the object come from `hru-lte.con`; see the connection file format.

`hru-lte.hru` is read only when the file exists and is not named `null`. When `lhru` is above zero, the file must be present: without it the HRU-lte objects have no properties and the run fails. A project with HRU-lte objects also needs `soils_lte.sol` (texture classes), `plants.plt` (plant parameters) and `lum.dtl` (the growing-season decision tables).

hru-lte.hru

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one HRU-lte with 35 values in the order below. The reader does not match columns by header name, so all 35 values must be present and in this order. Text values must not contain spaces.

#	Column	Units	Typical range	Description
1	`id`	-	1 and up	Row number. The `hlt` column of `hru-lte.con` refers to the row by this number. Ids must be unique; the rows may be in any order.
2	`name`	-	-	Name of the HRU-lte, up to 16 characters. Not used in calculations; the names in the output files come from `hru-lte.con`.
3	`area`	km² (in the source)	-	Drainage area. Used only to estimate the time of concentration when `t_conc` is zero, which does not work in 62.0.1 (see Time of concentration). The SWAT+ Editor writes the HRU area in hectares here.
4	`cn2`	-	35 to 98	SCS curve number for moisture condition II. See Surface runoff.
5	`cn3_swf`	fraction	0 to 0.95	Soil water factor for the curve number: the soil water content at which the retention parameter reaches its condition III value, as a fraction of the way from field capacity (0) to saturation (1). A value of 1 makes the shape coefficients of the retention curve infinite; use a value below 1. The SWAT+ Editor writes 0.
6	`t_conc`	min	10 to 1000	Time of concentration. Sets the peak runoff rate for erosion, and the delivery ratio when the HRU-lte is an element of a routing unit with a calculated delivery ratio. Give a value above zero. The SWAT+ Editor writes 26.
7	`soil_dp`	mm	300 to 3000	Depth of the soil profile. Multiplies the available water capacity and porosity of the soil texture.

#	Column	Units	Typical range	Description
8	`perc_co`	fraction	0 to 1	Percolation threshold: percolation from the soil profile starts when soil water exceeds `perc_co` times the available water capacity. Called `perco_co` in earlier documentation, which gave a range of 0 to 6000. See Percolation.
9	`slp`	m/m	0 to 0.6	Average land surface slope. Used for lateral flow and the USLE slope factor.
10	`slp_len`	m	10 to 150	Slope length. Used for lateral flow and the USLE slope length factor.
11	`et_co`	-	0.8 to 1.2	Evapotranspiration coefficient. Multiplies potential evapotranspiration, and again actual evapotranspiration. Values below 0.4 make evapotranspiration outside the growing season negative. See Evapotranspiration.
12	`aqu_sp_yld`	-	0.01 to 0.3	Specific yield of the shallow aquifer. Read but not used.
13	`alpha_bf`	1/day	0 to 1	Baseflow recession constant of the groundwater store. See Groundwater.
14	`revap`	fraction	0 to 0.2	Revap coefficient: water removed from the shallow groundwater store each day as a fraction of the day's actual evapotranspiration.
15	`rchg_dp`	fraction	0 to 1	Fraction of the day's percolation that moves from the shallow store to the deep store.
16	`sw_init`	fraction	0 to 1	Initial soil water as a fraction of the available water capacity.
17	`aqu_init`	mm	-	Initial shallow aquifer storage. Read but not used: the store starts at zero.
18	`aqu_sh_flo`	mm	-	Initial groundwater flow. Read but not used: the flow starts at zero.
19	`aqu_dp_flo`	mm	-	Initial deep aquifer storage. Read but not used: the store starts at zero.
20	`snow_h2o`	mm H_2O	-	Initial snow water content. Read but not used: the snowpack starts at zero.
21	`lat`	degrees	-90 to 90	Latitude. Read, and used only in a day length calculation whose result is not used, so it has no effect.
22	`soil_text`	-	-	Soil texture class, matched against the `texture` column of `soils_lte.sol`. See Name matching.
23	`trop_flag`	-	`non_trop`, `trop`	Tropical flag. Read but not used.
24	`grow_start`	-	-	Name of the `lum.dtl` decision table that starts the growing season, usually `pl_grow_sum` (or `pl_grow_win` for winter crops).
25	`grow_end`	-	-	Name of the `lum.dtl` decision table that ends the growing season and computes the yield, usually `pl_end_sum` (or `pl_end_win`).
26	`plnt_typ`	-	-	Plant name, matched against the `name` column of `plants.plt`.

#	Column	Units	Typical range	Description
27	`stress`	fraction	0 to 1	Plant stress factor for stresses the HRU-lte does not simulate (pests, root restriction, soil quality, nutrients). It multiplies daily biomass growth; 1 means no such stress. With 0 no biomass grows and the yield is zero. The SWAT+ Editor writes 0.
28	`pet_flag`	-	`harg`, `p_t`	Potential evapotranspiration method: `harg` for Hargreaves; any other value selects Priestley-Taylor.
29	`irr_flag`	-	-	Irrigation switch. See Irrigation for the values that switch irrigation on in 62.0.1.
30	`irr_src`	-	`outside_bsn`, `shal_aqu`, `deep_aqu`	Source of irrigation water: the HRU-lte's shallow or deep groundwater store, or an unlimited source outside the basin (any value other than `shal_aqu` and `deep_aqu`).
31	`t_drain`	hr	0, or 24 to 72	Time to drain the soil to field capacity through tile drains. 0 means no tile drains.
32	`usle_k`	0.013 t m² h/(m³ t cm)	0 to 0.65	USLE soil erodibility factor K.
33	`usle_c`	-	0.001 to 0.5	USLE cover and management factor C. It is constant through the year. The SWAT+ Editor writes the `usle_c_min` of the plant.
34	`usle_p`	-	0 to 1	USLE support practice factor P.
35	`usle_ls`	-	-	USLE slope length and steepness factor LS. Read but not used: SWAT+ computes LS from `slp` and `slp_len`.

Name matching

The text columns are compared character by character with the names in the other files. The comparison is case-sensitive and trailing blanks are ignored.

- `soil_text` is looked up among the twelve texture classes of `soils_lte.sol` (see the soils_lte.sol format). The matching class gives the available water capacity and porosity per mm of soil depth and the saturated hydraulic conductivity. If no class matches, the HRU-lte has no soil water capacity, and the run fails or produces invalid numbers when it divides by it. The SWAT+ Editor datasets contain the classes `sand`, `loamy_sand`, `sandy_loam`, `loam`, `silt_loam`, `silt`, `silty_clay_loam`, `clay_loam`, `sandy_clay_loam`, `sandy_clay`, `silty_clay` and `clay`, and the Editor picks one for each HRU-lte from the sand, silt and clay content of the first soil layer.
- `plnt_typ` is looked up in the `name` column of `plants.plt`. An unknown name writes `<name> not found (plants.plt)` to `diagnostics.out`, and the HRU-lte grows with the built-in default plant record, whose leaf area curve is not set up.
- `grow_start` and `grow_end` are looked up by table name in `lum.dtl`. An unknown name writes `<name> entry in (hru-lte.hru) not found in (lum.dtl)` to `diagnostics.out`; the run then refers to a decision table that does not exist, so both names must be present in `lum.dtl`.
- `pet_flag` is compared with `harg` only.
- `irr_flag` and `irr_src` are compared as described under Irrigation.

Daily water balance

Each HRU-lte runs once a day, in the order of the object list. The steps below follow that order. SW is the soil water content (mm), AWC and POR are the available water capacity and the porosity of the profile (mm), and

K_{sat} is the saturated hydraulic conductivity of the texture class (mm/h):

$$AWC = awc \cdot \text{soil_dp}, \qquad POR = por \cdot \text{soil_dp}, \qquad SW_0 = \text{sw_init} \cdot AWC$$

where *awc* and *por* are the per-mm values of the texture class in `soils_lte.sol`.

Snow

When the mean air temperature $\bar{T}$ is below 0 °C, the whole day's precipitation is added to the snowpack and there is no runoff or infiltration that day. Otherwise snowmelt is

$$SNO_{mlt} = 4.57\,\bar{T}$$

limited to the snowpack, and rain and melt together form the water input P for the day. The `snow.sno` parameters are not used by HRU-lte objects.

Surface runoff

Surface runoff uses the SCS curve number method with a retention parameter that varies with soil water (see SCS curve number). At the start of the run the condition I and III curve numbers are derived from `cn2`:

$$CN_1 = CN_2 - \frac{20\,(100 - CN_2)}{100 - CN_2 + \exp\,(2.533 - 0.063\,(100 - CN_2))}, \qquad CN_1 \geq 0.4\,CN_2$$

$$CN_3 = CN_2 \exp\,(0.00673\,(100 - CN_2))$$

$$S_{max} = 254\left(\frac{100}{CN_1} - 1\right), \qquad S_3 = 254\left(\frac{100}{CN_3} - 1\right)$$

Each day the retention parameter is

$$S = S_{max}\left(1 - \frac{SW}{SW + \exp(w_1 - w_2\,SW)}\right), \qquad S \geq 3$$

The shape coefficients w_1 and w_2 are fixed at the start so that $S = S_3$ when $SW = AWC + \text{cn3_swf}\,(POR - AWC)$, and $S = 2.54$ when $SW = POR$. Runoff is

$$Q_{surf} = \frac{(P - 0.2\,S)^2}{P + 0.8\,S} \quad \text{for } P > 0.2\,S$$

and zero otherwise. The rest of P is added to the soil water. SWAT+ also stores `cn2` limited to 35 to 98, but the runoff curve is built from the value as given in the file, so values outside that range are used as they are.

Evapotranspiration

Potential evapotranspiration is computed with the method chosen in `pet_flag` and multiplied by `et_co`. With Hargreaves,

$$PET = \text{et_co} \cdot 0.0032\,\frac{H_0}{2.5 - 0.0022\,\bar{T}}\,(\bar{T} + 17.8)\,(T_{max} - T_{min})^{0.6}$$

where H_0 is the day's maximum possible solar radiation (MJ/m²). With Priestley-Taylor,

$$PET = \text{et_co} \cdot 1.28 \, \frac{\Delta}{\Delta + 0.68} \cdot \frac{23.9 \, H \, (1 - 0.23)}{58.3}, \qquad \Delta = \frac{5304}{T_K^2} \exp\left(21.255 - \frac{5304}{T_K}\right)$$

where H is the day's solar radiation (MJ/m²) and T_K the mean air temperature in kelvin. Both are simpler forms than those used for ordinary HRUs (see Potential evapotranspiration).

Actual evapotranspiration is proportional to the soil water content:

$$E_a = b \cdot PET \, \frac{SW}{AWC}, \qquad b = \begin{cases} \text{et_co} & \text{during the growing season} \\ \text{et_co} - 0.4 & \text{outside it} \end{cases}$$

It is taken from the soil after lateral flow, tile flow and percolation, and is limited to the soil water left at that point. Because `et_co` enters both PET and b, a change in `et_co` changes actual evapotranspiration roughly with its square during the growing season.

Lateral flow

When soil water is above the available water capacity,

$$Q_{lat} = 0.024 \, \frac{SW - AWC}{POR - AWC} \, K_{sat} \, \frac{\text{slp}}{\text{slp_len}}$$

limited to the soil water.

Tile flow

When `t_drain` is above zero and soil water is above the available water capacity,

$$Q_{tile} = (SW - AWC) \left(1 - \exp\left(\frac{-24}{\text{t_drain}}\right)\right), \qquad Q_{tile} \leq 10 \text{ mm}$$

Percolation

Water above `perc_co` times the available water capacity drains from the bottom of the profile:

$$w_{perc} = (SW - \text{perc_co} \cdot AWC) \left(1 - \exp\left(\frac{-24}{TT}\right)\right), \qquad TT = \max\left(2, \; \frac{POR - AWC}{K_{sat}}\right)$$

where TT is the travel time in hours. With `perc_co` at 1 only water above field capacity percolates; with `perc_co` at 0 (the value the SWAT+ Editor writes) the profile can drain well below field capacity.

Groundwater

The shallow and deep groundwater stores of an HRU-lte are separate from the aquifer objects. Percolation is added to the shallow store. Groundwater flow follows a recession with $\alpha =$ `alpha_bf`:

$$Q_{gw,d} = Q_{gw,d-1} \, e^{-\alpha} + w_{perc} \, (1 - e^{-\alpha})$$

limited to the shallow storage. Then revap ($\text{revap} \cdot E_a$) and deep percolation ($\text{rchg_dp} \cdot w_{perc}$) are taken from what is left in the shallow store, each limited to the storage. Deep percolation is added to the deep store, which only irrigation from `deep_aqu` draws on. Revap leaves the HRU-lte; it is not added to the soil or to the reported evapotranspiration. Both stores and the groundwater flow start at zero.

Water yield

The water the HRU-lte sends downstream each day is

$$WYLD = Q_{surf} + Q_{lat} + Q_{tile} + Q_{gw}$$

In the `hru-lte_wb` output files, `latq` is the sum of lateral flow and groundwater flow.

Plant growth

The plant named in `plnt_typ` grows between the two decision tables. On each day outside the growing season the `grow_start` table is checked; when its `grow_init` action runs, growth starts with zero biomass and leaf area. During the season the `grow_end` table is checked; its `grow_end` action computes the yield and ends the season. The `pl_grow_sum` and `pl_end_sum` tables of the SWAT+ reference data start growth when the fraction of the year's base-zero heat units at the weather station is between 0.15 and 0.2, and end it when the fraction is above 0.5 or on day 330. See decision table actions.

Heat units to maturity

At the start of the run SWAT+ estimates the heat units to maturity, PHU, from the long-term monthly mean temperatures in the weather generator of the HRU-lte's station. Daily values of $\bar{T} - T_{base}$ above zero are summed over the days between a start and an end day, and

$$PHU = \max(500,\ 0.9 \sum(\bar{T} - T_{base}))$$

limited to 2000 for annual plant types (`warm_annual`, `cold_annual` and their tuber forms). In 62.0.1 the start and end days of the sum are the positions of the `grow_start` and `grow_end` tables in `lum.dtl` (first table 1, second table 2, and so on). The days of the year of the growing season are not used. For two tables next to each other in `lum.dtl` the sum covers no days and PHU takes its minimum of 500.

Biomass and leaf area

During the growing season the fraction of heat units g increases each day by $(\bar{T} - T_{base})/PHU$. Biomass grows with intercepted radiation:

$$\Delta B = \gamma \cdot \text{stress} \cdot bm_e \cdot 0.5\, H \left(1 - e^{-0.65\,(LAI+0.05)}\right), \qquad \gamma = \min(ws,\ ts,\ as)$$

where bm_e is `bm_e` of `plants.plt`, $ws = E_a/PET$ is the water stress factor (before the adjustment b), ts the temperature stress factor and as the aeration stress factor:

$$ts = \exp\left(-0.1054 \left(\frac{T_{opt} - \bar{T}}{2T_{opt} - T_{base} - \bar{T}}\right)^2\right) \text{ for } \bar{T} > T_{base}, \qquad ts = 0 \text{ otherwise}$$

Aeration stress starts when SW exceeds AWC by more than 85% of the drainable range $POR - AWC$. Leaf area follows the leaf area curve of the plant (`frac_hu1`, `lai_max1`, `frac_hu2`, `lai_max2` and `lai_pot` in `plants.plt`), with the daily increase scaled by $\sqrt{\gamma}$. See plant growth for the background of these relations.

Yield

When the season ends, the yield is

$$Y = 0.8\, B\, HI', \qquad HI' = (HI - HI_{min}) \frac{wur}{wur + \exp(6.13 - 0.0883\, wur)} + HI_{min}$$

where HI is `harv_idx`, HI_{min} is `harv_idx_ws` of `plants.plt`, and wur is 100 times the ratio of actual to potential evapotranspiration summed over the season (100 when the potential sum is below 10 mm). Biomass and leaf area are then set to zero.

Plant growth has no effect on the water balance or on erosion: evapotranspiration does not depend on leaf area, and `usle_c` is constant. The plant affects the water balance only through the growing season itself (the factor b) and through irrigation.

Irrigation

During the growing season, on days when the water stress factor ws is below 0.8, the soil is filled to the available water capacity with at most 76.2 mm. With `irr_src` set to `shal_aqu` or `deep_aqu` the water is taken from the HRU-lte's shallow or deep store and limited to what it contains; any other value takes it from an unlimited source outside the basin.

In 62.0.1 irrigation is switched on when `irr_flag` sorts alphabetically after `no_irr`. The value `irr`, listed as the "irrigation" value in earlier documentation, sorts before `no_irr` and so leaves irrigation off. Values such as `yes` or `sprinkler` switch it on.

Erosion

Sediment yield is computed with MUSLE (see MUSLE). The slope length and steepness factor is computed at the start of the run from `slp` and `slp_len`; the `usle_ls` column is not used:

$$LS = \left(\frac{\text{slp_len}}{22.128}\right)^{m} \left(65.41\,\sin^2\theta + 4.56\,\sin\theta + 0.065\right), \qquad m = 0.6\left(1 - e^{-35.835\,\text{slp}}\right)$$

with $\theta = \arctan(\text{slp})$. Sediment from surface runoff is

$$sed = 11.8\,(Q_{surf}\; q_{peak} \cdot 1000\,A)^{0.56}\; K\,C\,P\,LS$$

with K, C and P from `usle_k`, `usle_c` and `usle_p`, and A the `area` of `hru-lte.con`. A concentration of 500 mg/L in lateral and groundwater flow is added to this. There is no snow cover or residue effect.

Time of concentration

The peak runoff rate uses the time of concentration t_c in seconds (`t_conc` times 60):

$$q_{peak} = \frac{2\,Q_{surf} \cdot 1000\,A}{1.5\,t_c} + \frac{(Q_{lat} + Q_{gw}) \cdot 1000\,A}{86400}$$

When `t_conc` is zero, SWAT+ tries to estimate it with the Kirpich formula, using `area` as the channel length. The slope in that formula is always zero in 62.0.1, so the estimate is infinite: depending on how the model was compiled the run either stops with a floating-point error or continues with no surface runoff in the peak rate. Give `t_conc` a value above zero.

Area units

The HRU-lte calculations convert a depth D in mm to a volume with $1000\,D\,A$ and report sediment yield per $100\,A$, which are the conversions for an area A in km². A is the `area` column of `hru-lte.con`, which is in hectares for every other object type and which the SWAT+ Editor writes in hectares. With hectares, the flow volumes and sediment loads an HRU-lte sends downstream are 100 times the depth over its area, and the peak rate and sediment yield in the output are affected in the same way. Only the total hydrograph is filled: an HRU-lte connection in `hru-lte.con` that asks for the surface, lateral, tile or recharge hydrograph sends nothing.

Calibration

Parameters with the suffix `_lte` in `calibration.cal` change `hru-lte.hru` values (see simplified HRU parameters). The changes are applied after the file has been read and the HRU-lte objects have been set up, so in 62.0.1 they have an effect only for values the daily calculation reads directly from the file:

- effective: `awc_lte` (available water capacity of the object), `slope_lte` and `slopelen_lte` (lateral flow only), `revapc_lte`, `percc_lte`, `abf_lte` and `tc_lte`
- no effect: `cn2_lte`, `etco_lte`, `soildep_lte`, `sw_lte`, `sy_lte`, `gw_lte`, `gwflow_lte`, `gwdeep_lte`, `snow_lte`, `uslek_lte`, `uslec_lte`, `uslep_lte` and `uslels_lte`

`abf_lte` changes the daily recession factor $e^{-\alpha}$ in place of `alpha_bf`, and `tc_lte` changes the time of concentration in seconds. Except for `awc_lte`, a change to object number n is applied to the `hru-lte.hru` row with `id` n; the two agree when row numbers follow the objects, as the SWAT+ Editor writes them.

Example

No `hru-lte.hru` is included in the SWAT+ reference data. The file below is illustrative, with the column headers the SWAT+ Editor writes.

`hru-lte.hru: illustrative example`

id	name	area	cn2	cn3_swf	t_conc	soil_dp	perc_co	slp	slp_len	et_co	aqu_sp_yld	alpha_bf	rev
1	hlt001	52.3	77.0	0.00	26.0	1500.0	1.00	0.04	60.0	1.00	0.05	0.05	0.

One HRU-lte with a 1.5 m loam profile that starts half full. Percolation starts above field capacity (`perc_co` 1). Corn grows between the `pl_grow_sum` and `pl_end_sum` tables with no extra stress (`stress` 1), there is no irrigation and no tile drainage, and potential evapotranspiration follows Hargreaves. The values of `area`, `aqu_sp_yld`, `aqu_init`, `aqu_sh_flo`, `aqu_dp_flo`, `snow_h2o`, `lat`, `trop_flag` and `usle_ls` have no effect in 62.0.1.

Related files

- Connection files: `hru-lte.con` defines the HRU-lte objects, their area, weather station and outflow connections; its `hlt` column selects a row of this file by `id`.
- Object counts: the `lhru` column of `object.cnt` gives the number of HRU-lte objects.
- Master file: `file.cio` names this file on its `hru` line, and `soils_lte.sol` on its `soils` line.
- Soil texture classes: `soils_lte.sol` supplies available water capacity, porosity and conductivity for `soil_text`.
- Plant parameters: `plants.plt` supplies the plant parameters for `plnt_typ`.
- Decision tables: the `grow_start` and `grow_end` tables in `lum.dtl`.
- Weather files: the weather station named in `hru-lte.con` supplies daily weather and, through its weather generator, the temperatures for the heat units to maturity.
- Water balance output: `hru-lte_wb` files; plant growth and stress go to the `hru-lte_pw` files.

Sources

- Reader: `hru_lte_read.f90`, called from `main.f90.in` after `dtbl_lum_read`
- Type definitions: `type swatdeg_hru_data` and `type swatdeg_hru_dynamic` in `hru_lte_module.f90`
- Daily water balance, plant growth, irrigation and erosion: `hru_lte_control.f90`, called from `command.f90`
- Retention curve shape: `ascrv.f90`
- Growing season actions `grow_init` and `grow_end`: `actions.f90`; conditions with `obj hlt`: `conditions.f90`
- Texture classes: `soil_lte_db_read.f90`, `type soil_lte_database` in `soil_data_module.f90`
- Plant parameters: `plant_data_module.f90`, `plant_parm_read.f90`
- Connection and object numbering: `hyd_connect.f90`, `hyd_read_connect.f90`
- Delivery ratio from `t_conc`: `dr_ru.f90`
- Calibration: `cal_parm_select.f90`, `cal_parmchg_read.f90`, `proc_cal.f90`
- Output: `hru_lte_output.f90`, `output_landscape_init.f90`
- File name slot: `in_hru%hru_ez` in `input_file_module.f90`, read in `readcio_read.f90`

Column	Fortran variable
id	row index of hlt_db
name	hlt_db%name
area	hlt_db%dakm2
cn2	hlt_db%cn2
cn3_swf	hlt_db%cn3_swf
t_conc	hlt_db%tc
soil_dp	hlt_db%soildep
perc_co	hlt_db%perco
slp	hlt_db%slope
slp_len	hlt_db%slopelen
et_co	hlt_db%etco
aqu_sp_yld	hlt_db%sy
alpha_bf	hlt_db%abf
revap	hlt_db%revapc
rchg_dp	hlt_db%percc
sw_init	hlt_db%sw
aqu_init	hlt_db%gw
aqu_sh_flo	hlt_db%gwflow
aqu_dp_flo	hlt_db%gwdeep
snow_h2o	hlt_db%snow
lat	hlt_db%xlat
soil_text	hlt_db%text
trop_flag	hlt_db%tropical
grow_start	hlt_db%igrow1
grow_end	hlt_db%igrow2
plnt_typ	hlt_db%plant
stress	hlt_db%stress
pet_flag	hlt_db%ipet
irr_flag	hlt_db%irr
irr_src	hlt_db%irrsrc
t_drain	hlt_db%tdrain
usle_k	hlt_db%uslek
usle_c	hlt_db%uslec
usle_p	hlt_db%uslep
usle_ls	hlt_db%uslels

Purpose

A routing unit collects the outflow of a group of objects, usually the HRUs of one landscape unit, and routes the sum as a single object. Its row in rout_unit.con sends the result on, typically the total flow to a channel and the recharge to an aquifer. Three files describe routing units: rout_unit.def lists the elements of each unit, rout_unit.ele describes each element, and rout_unit.rtu gives each unit its topography and field records.

A landscape unit is a group of HRUs used only for output. ls_unit.def lists the elements of each landscape unit and ls_unit.ele gives each element its area fractions. The same element file also weights the HRUs in the basin output files. aqu_catunit.ele lists aquifer elements; in 62.0.1 its only effect is to switch on the data rows of the basin aquifer output.

The files are named in file.cio:

File	file.cio line	Slot	Read
rout_unit.def	routing_unit (line 8)	1	when the rtu count in object.cnt is above 0
rout_unit.ele	routing_unit (line 8)	2	when the rtu count is above 0
rout_unit.rtu	routing_unit (line 8)	3	when the rtu count is above 0
ls_unit.ele	regions (line 26)	1	every run

File	file.cio line	Slot	Read
ls_unit.def	regions (line 26)	2	every run
aqu_catunit.ele	regions (line 26)	9	every run, with the calibration inputs

Slot 4 of the routing_unit line (rout_unit.dr) is read from file.cio but never opened.

All six files are free format and whitespace-separated. The readers take the values in column order and do not check the header names.

For rout_unit.rtu, a name of null or a name that does not exist in the run folder means the file is not read. The other five files are skipped when the name is null. When the name is not null and the file does not exist, SWAT+ creates an empty file of that name in the run folder, reads nothing from it, and continues as if the name were null. What a missing file means for the run is given under each file.

rout_unit.def

Format

Line 1 is a title and line 2 is the column header; both are skipped. Each later line defines one routing unit. The line ends with a list of element numbers, and only the first number of that list has a header.

#	Column	Units	Description
1	id	-	Number of the routing unit. Not used: the rows are taken in order, so row n defines routing unit n.
2	name	-	Name of the routing unit. Not used.
3	elem_tot	-	Number of values in the element list that follows. When the list uses ranges, this is smaller than the number of elements.
4 and on	elements	-	Element list: id values of rows in rout_unit.ele, written as described under Element lists.

Element lists

rout_unit.def and ls_unit.def use the same list format. Each value is an element number, and a negative value closes a range that starts at the value before it. The list 1 -4 7 therefore means elements 1, 2, 3, 4 and 7, and elem_tot is 3 because the list has three values. A list with a single value gives one element. An elem_tot of 0 gives a unit with no elements.

A negative value must follow a positive one. Two negative values in a row, or a negative first value in a list of more than one value, put negative element numbers into the list.

Behaviour

- Routing unit n is the routing unit whose id in rout_unit.con is n. It uses row n of rout_unit.def, counted from the first data line, whatever the id column says. SWAT+ reads as many rows as the rtu count in object.cnt. Extra rows are ignored, and units without a row have no elements.
- The define column of rout_unit.rtu is not used to find the row. The routing unit ids in rout_unit.con, the rows of rout_unit.def and the ids in rout_unit.rtu therefore have to run 1, 2, 3 in the same order. The SWAT+ Editor writes them that way.
- An object can be an element of one routing unit only. If two element rows of rout_unit.ele point to the same object and both are listed in rout_unit.def (in one unit or in two), SWAT+ stops at start-up with an allocation error.
- A routing unit with no elements produces no outflow, and no message is written.
- If rout_unit.def is missing or null in a project with routing units, every routing unit has no elements. The HRUs then reach the rest of the model only through their rows in hru.con, and an HRU without outflow connections there loses its water.

Example

`refdata/0su_1hru/rout_unit.def` in the SWAT+ repository:

```
rout_unit.def: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id              name  elem_tot  elements
       1             rtu001         1         1
```

Routing unit 1 has one element, element 1 of `rout_unit.ele`.

rout_unit.ele

Format

Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one element.

#	Column	Units	Description
1	`id`	-	Element number, used in the element lists of `rout_unit.def`. The rows can be in any order and the numbers do not have to be consecutive; the largest `id` sets the size of the table.
2	`name`	-	Name of the element. Not used.
3	`obj_typ`	-	Object type of the element, one of the codes below.
4	`obj_id`	-	Number of the object within its type: its `id` in the connection file of that type.
5	`frac`	fraction	Fraction of the element in the routing unit, used to scale the element's outflow. See Expansion factor.
6	`dlr`	-	Delivery ratio: `0` or `calc`, `1` or `full`. See Delivery ratio.

The object type codes SWAT+ recognises are:

`obj_typ`	Object	Connection file
`hru`	HRU	`hru.con`
`hlt`	HRU-lte	`hru-lte.con`
`ru`	routing unit	`rout_unit.con`
`cha`	legacy channel	`channel.con`
`sdc`	SWAT+ channel	`chandeg.con`
`res`	reservoir	`reservoir.con`
`exc`	export coefficient	`exco.con`
`dr`	delivery ratio	`delratio.con`
`out`	outlet	`outlet.con`

Any other code, including `aqu`, `rec` and `rtu`, is not recognised. No message is written, and the element is linked to object 1, the first object in the model (HRU 1 in a model with HRUs). Earlier documentation listed aquifers, recall objects and "another rtu" as possible elements; in 62.0.1 a routing unit inside a routing unit has the code `ru`.

Expansion factor

Each day a routing unit adds up the hydrographs of its elements. Each element's hydrographs are multiplied by an expansion factor e before they are added:

- HRU and HRU-lte elements with `frac` below 0.99999:

$$e = \text{frac}\ \frac{A_{ru}}{A_{obj}}$$

  where A_{ru} is the routing unit's `area` in `rout_unit.con` and A_{obj} the HRU's `area` in its connection file. When `frac` is the HRU's share of the routing unit area, e is 1 and the whole HRU outflow is added.

- HRU and HRU-lte elements with `frac` of 0.99999 or more: $e = 1$, so the whole HRU outflow is added whatever the areas.
- Export coefficient elements: e as for HRUs when the export coefficient object has an area above 0.01 ha, otherwise 1.
- All other elements: $e =$ frac, the fraction of the object's outflow that enters the routing unit.

The `frac` values are not checked. If they do not add up as intended, the routing unit outflow is scaled accordingly.

The factor and the delivery ratio apply to the total, surface, lateral and tile hydrographs. The recharge hydrograph is multiplied by the factor only. Pesticide, salt and constituent loads are added in full: 62.0.1 applies neither the expansion factor nor the delivery ratio to them.

When another object sends flow to a routing unit through its connection file, the inflow is shared among the unit's elements. Each element receives the sender's fraction multiplied by its `frac`. How an HRU uses that inflow is described under Inflow to HRUs and routing units.

Delivery ratio

The `dlr` column sets a ratio for each hydrograph variable of the element:

`dlr`	Ratio
`0` or `calc`	Flow 1. All other variables (sediment, nutrients, chlorophyll-a, CBOD, dissolved oxygen, sediment size classes) are multiplied by a calculated ratio, given below.
`1` or `full`	1 for every variable.
any other value	0 for every variable.

For HRU and HRU-lte elements the calculated ratio is

$$DR = \min\left(1,\ \sqrt{\frac{t_{c,\mathrm{elem}}}{t_{c,ru}}}\right)$$

where $t_{c,\mathrm{elem}}$ is the HRU's time of concentration (see Surface runoff lag), or for an HRU-lte the time of concentration of its `hru-lte.hru` record, and $t_{c,ru}$ is the routing unit's time of concentration (see Time of concentration). For `sdc` and `ru` elements the calculated ratio is 1. For the other element types no ratio is calculated, and the element takes the ratio last calculated for an earlier element (0 if there is none).

A `dlr` value that names a record in `delratio.del` does not work in 62.0.1. The names are looked up when the routing units are set up, and `delratio.del` is read only after that, so no name is ever found. The ratio then stays at 0 for every variable, and the element contributes only its recharge hydrograph. The same happens for any other value, such as `null`. Use `0`, `calc`, `1` or `full`. The SWAT+ Editor writes `0`.

The decision table action `hru_fr_update` (see decision tables) reads a replacement `rout_unit.ele` during the run. By then `delratio.del` has been read, so a `dlr` that names a `delratio.del` record does take that record's ratios from `dr_om.del`. Elements whose `dlr` is `0`, `calc`, `1`, `full` or an unknown name keep the ratios they had before.

Behaviour

- `rout_unit.ele` is required in a project with routing units. If it is missing or `null`, the routing unit tables are never set up and the run fails during start-up.
- In sub-daily runs (time step below one day) with Green and Ampt infiltration (`gampt` = 1 in `codes.bsn`), the routing unit's sub-daily flow is set from each HRU element in turn, so only the last HRU element in the list is kept, and the result is shifted by a fixed 2 hours instead of the routing unit's time of concentration.
- The `ru` row of `print.prt` writes the sum of the element hydrographs of each routing unit to the `ru_*` files (see ru files).

Example

`refdata/Osu_1hru/rout_unit.ele`:

```
rout_unit.ele: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id  name                    obj_typ    obj_id          frac               dlr
       1  hru0001                     hru         1             1                 0
```

Element 1 is HRU 1. With `frac` = 1 the whole HRU outflow enters the routing unit. With `dlr` = 0 the water is delivered in full, and sediment and nutrients are multiplied by the calculated delivery ratio.

rout_unit.rtu

Format

Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one routing unit.

#	Column	Units	Description
1	`id`	-	Number of the routing unit. Routing unit n (the `id` in `rout_unit.con`) uses the row whose `id` is n. The ids must lie between 1 and the `rtu` count in `object.cnt`. An `id` of 0 ends the file.
2	`name`	-	Name of the routing unit. Not used.
3	`define`	-	Name of the `rout_unit.def` row. Not used: routing unit n always uses row n of `rout_unit.def`.
4	`dlr`	-	Delivery ratio of the routing unit. Not used; delivery ratios are set per element in `rout_unit.ele`.
5	`topo`	-	Name of a record in `topography.hyd`. Its slope and slope length are used for the routing unit's time of concentration.
6	`field`	-	Name of a record in `field.fld`. Its field length is used for the routing unit's time of concentration.

Time of concentration

The routing unit's time of concentration is the sum of an overland flow time and a channel flow time:

$$t_{ov} = \frac{0.0556\,(L_{slp}\,n)^{0.6}}{(slp + 0.001)^{0.3}}, \qquad t_{ch} = \frac{0.62\,L\,n^{0.75}}{A^{0.125}\;(0.5\,(slp + 0.001))^{0.375}}, \qquad t_{c,ru} = t_{ov} + t_{ch}$$

Symbol	Meaning	Units
t_{ov}, t_{ch}, $t_{c,ru}$	overland, channel and total time of concentration	hr
L_{slp}	slope length, `slp_len` of the `topo` record	m
slp	average slope, `slp` of the `topo` record	m/m
L	channel length, `len` of the `field` record divided by 1000	km
A	routing unit area, `area` in `rout_unit.con`	km²
n	Manning's n for overland flow, averaged over the HRU elements	-

The Manning's n is the area-weighted mean of the `ov_mann` values of the HRU elements' land uses (see `landuse.lum`):

$$n = \frac{\sum_i n_i\,A_i}{A}$$

where A_i is the full area of HRU element i; `frac` is not used. When the routing unit has an element that is not an HRU, the running sum is set to 0.1 at that element and the HRU elements after it are added to 0.1. The result then depends on the order of the elements.

The time of concentration is used for the calculated delivery ratio of the elements and, in sub-daily runs, for the unit hydrograph of the routing unit.

Behaviour

- A `topo` name that is `null` or not found in `topography.hyd` gives the first record of that file. A `field` name that is `null` or not found gives the built-in field length of 500 m. No message is written in either case.
- `rout_unit.rtu` is required in a project with routing units. If it is missing or `null`, no routing unit records are set up and the run fails during start-up.

Example

`refdata/Osu_1hru/rout_unit.rtu`:

```
rout_unit.rtu: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id              name            define               dlr              topo             field
       1            rtu01             rtu001              null       toportu011            fld011
```

`fld011` is in `field.fld` with a length of 500 m. `toportu011` is not in this dataset's `topography.hyd`, so the routing unit uses its first record, `topohru0001` (slope 0.01078, slope length 10 m), without a message. `define` and `dlr` have no effect.

ls_unit.def

Format

Line 1 is a title, line 2 gives the number of landscape units, and line 3 is the column header. Each later line defines one landscape unit, and the line ends with an element list as in `rout_unit.def`.

#	Column	Units	Description
1	`id`	-	Number of the landscape unit. Not used: row n defines landscape unit n.
2	`name`	-	Name of the landscape unit, written in the `name` column of the landscape unit output files.
3	`area`	ha	Area of the landscape unit. Not used.
4	`elem_tot`	-	Number of values in the element list. 0 gives a landscape unit with no elements, whose output is all zeros.
5 and on	`elements`	-	Element list: `id` values of rows in `ls_unit.ele`, written as described under Element lists.

SWAT+ reads as many rows as line 2 gives.

Behaviour

- Landscape units are used for output only. Each one gets a row in the `lsunit_wb`, `lsunit_nb`, `lsunit_ls` and `lsunit_pw` files and in the landscape unit carbon files (`lsu_cb_gl`, `lsu_cb_trf`, `lsu_cb_plt`), switched on in `print.prt`. Each value is the sum over the unit's elements of the HRU (or HRU-lte) value multiplied by the element's `sub_frac` in `ls_unit.ele`. See Aggregation.
- The element number is used as the HRU number: element n adds the output of HRU n (or HRU-lte n), and its `obj_typ_no` is not consulted. The element ids in `ls_unit.ele` therefore have to equal the HRU numbers.
- In the output files, `unit` is the landscape unit number and `gis_id` is taken from the routing unit with the same number in `rout_unit.con`.
- With gwflow and landscape unit connection (`conn_type` 2, see `lsucell.gw`), the element lists also decide which HRUs pass recharge and remaining evapotranspiration to the cells of each landscape unit. Here too element n is HRU n.
- `ls_unit.def` needs `ls_unit.ele`. If `ls_unit.def` is missing or `null`, the project has no landscape units and the landscape unit output files are not opened.

Example

`refdata/Osu_1hru/ls_unit.def`:

```
ls_unit.def: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
1
      id              name          area  elem_tot  elements
       1            rtu001       10.0000         1         1
```

One landscape unit of 10 ha, named `rtu001` after the routing unit, with element 1 of `ls_unit.ele`.

ls_unit.ele

Format

Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one element. Six values are read from each line.

#	Column	Units	Description
1	`id`	-	Element number, used in the element lists of `ls_unit.def`. The largest `id` sets the size of the table. Must equal the HRU number (see below).
2	`name`	-	Name of the element. Not used.
3	`obj_typ`	-	`hru` or `hlt`. Elements of any other type are skipped in the sums.
4	`obj_typ_no`	-	HRU or HRU-lte number.
5	`bsn_frac`	fraction	Fraction of the basin land area (`ls_area` in `object.cnt`) covered by the HRU. Weight of the HRU in the basin output files.
6	`sub_frac`	fraction	Fraction of the landscape unit area covered by the HRU. Weight of the HRU in the landscape unit output files. Called `lsu_frac` in earlier documentation; the `Ames_sub1` file labels it `RU_FRAC`.

Behaviour

- Basin output: the `basin_wb`, `basin_nb`, `basin_ls` and `basin_pw` files, including the starting soil water and snow in `basin_wb`, are the sums of the HRU values multiplied by `bsn_frac`. The fractions are used as given; if they do not add up to 1, the basin values are scaled by their sum.
- For each HRU number n, the basin sum reads row n, takes its `obj_typ_no` m, and then uses the `obj_typ` and `bsn_frac` of row m with the output of HRU m. The sum is right when row n describes HRU n, which is how the SWAT+ Editor writes the file. HRU-lte objects are summed the same way over the first rows of the file.
- Landscape unit output uses `sub_frac` (see `ls_unit.def`).
- The soft calibration of the whole basin in `water_balance.sft` takes each HRU's area as `ls_area` times its `bsn_frac` (see soft calibration).
- The decision table action `hru_fr_update` reads a replacement `ls_unit.ele` during the run and sets the area of each `hru` element to `bsn_frac` times `ls_area`.
- If `ls_unit.ele` is missing or `null`, the basin landscape files are opened and get their headers, and no data rows are written. The routine that stores the basin's starting soil water at the end of the warm-up years still reads the element table, so a model with HRUs and no `ls_unit.ele` can stop with a memory error at that point. Treat the file as required in any model with HRUs.

Example

`refdata/Ames_sub1/ls_unit.ele`:

```
lsu_unit.ele
          ID    NAME      OBJ_TYP  OBJ_TYP_NO    BSN_FRAC       RU_FRAC
           1    hru0001       hru           1     0.20           0.0
           2    hru0002       hru           2     0.20           0.0
```

```
 3    hru0003     hru          3     0.20          0.0
 4    hru0004     hru          4     0.20          0.0
 5    hru0005     hru          5     0.10          0.0
 6    hru0006     hru          6     0.10          0.0
 7    hru0007     hru          7     0.20          0.0
 8    hru0008     hru          8     0.20          0.0
 9    hru0009     hru          9     0.20          0.0
10    hru0010     hru         10     0.20          0.0
11    hru0011     hru         11     0.10          0.0
12    hru0012     hru         12     0.10          0.0
```

This file is not in the SWAT+ Editor layout: it has six columns and upper-case headers. Each row describes the HRU of the same number, as the basin sums require. The `BSN_FRAC` values add up to 2.0, so every basin output of this dataset is twice the area-weighted HRU average. `Ames_sub1` has no `ls_unit.def`, so the `RU_FRAC` values of 0 are not used.

aqu_catunit.ele

Format

Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one element, and seven values are read from it. A line with fewer values takes the missing ones from the next line.

#	Column	Units	Description
1	`id`	-	Element number. The largest `id` sets the size of the table.
2	`name`	-	Name of the element.
3	`obj_typ`	-	Object type, `aqu` in files written by the SWAT+ Editor.
4	`obj_typ_no`	-	Aquifer number.
5	`bsn_frac`	fraction	Fraction of the basin covered by the aquifer.
6	`sub_frac`	fraction	Fraction of the aquifer region covered by the element.
7	`reg_frac`	fraction	Fraction of the calibration region covered by the element.

Behaviour

- In 62.0.1 the file works as a switch. When it has at least one element, the basin aquifer files (`basin_aqu` row in `print.prt`) get one data row per interval. Without it they get headers only.
- The basin aquifer values are weighted by each aquifer's area divided by `tot_area` in `object.cnt` (see aquifer output). The columns of `aqu_catunit.ele` are not used for this.
- The columns are meant for aquifer regions defined in `aqu_catunit.def` and `aqu_reg.def`. SWAT+ 62.0.1 reads those files, but no calculation or output uses the regions.

Example

The first rows of `refdata/Osu_1hru/aqu_catunit.ele`:

```
aqu_catunit.ele: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id  name                    obj_typ  obj_typ_no      bsn_frac      sub_frac      reg_frac
       1  aqu011                      aqu           1       0.00241       0.00000       0.00000
       2  aqu012                      aqu           2       0.10460       0.00000       0.00000
       3  aqu021                      aqu           3       0.01290       0.00000       0.00000
```

The file has 21 elements, while the dataset has one aquifer. The rows have no effect beyond switching on the basin aquifer output. In `refdata/Ames_sub1` the file is empty, so that dataset writes no basin aquifer data rows.

Columns in earlier documentation

- `ls_unit.ele`: the SWAT+ Editor writes a seventh column, `reg_frac`. SWAT+ 62.0.1 reads six values per line and ignores it.
- `ls_unit.def`: earlier documentation did not list the line with the number of landscape units (line 2) or the `area` column. Both are read, and a file without them is read out of step.

Related files

- `rout_unit.con`: the routing unit objects, their areas and their outflow connections.
- `hru.con` and `hru-data.hru`: the HRUs that are elements of routing and landscape units.
- `object.cnt`: the `rtu` count, and `ls_area` and `tot_area` used with `bsn_frac` and the basin aquifer output.
- `file.cio`: names the files on the `routing_unit` and `regions` lines.
- `topography.hyd` and `field.fld`: the `topo` and `field` records of `rout_unit.rtu`.
- `aquifer.aqu`: the aquifers listed in `aqu_catunit.ele`.
- `print.prt`: switches for the `ru`, `lsunit_*`, `basin_*` and `basin_aqu` outputs.
- Decision tables: the `hru_fr_update` action.
- Outputs: water balance, nutrient balance and losses, hydrographs, aquifer output.
- Delivery ratios in the model theory.

Sources

- File name slots: `type input_ru` and `type input_regions` in `input_file_module.f90`, read in `readcio_read.f90`; read order in `main.f90.in` and `hyd_connect.f90`
- Routing unit readers: `ru_read.f90` (`rout_unit.rtu`), `ru_read_elements.f90` (`rout_unit.ele`, `rout_unit.def`); element lists: `define_unit_elements.f90`
- Routing unit types: `ru_parameters` in `ru_module.f90`, `routing_unit_data` and `routing_unit_elements` in `hydrograph_module.f90`
- Routing unit linkage and inflow sharing: `hyd_connect.f90`, `hyd_read_connect.f90`
- Daily routing unit sum: `ru_control.f90`, called from `command.f90`
- Delivery ratios: `dr_ru.f90`, `dr_db_read.f90`
- Time of concentration and unit hydrographs: `time_conc_init.f90`, `unit_hyd_ru_hru.f90`
- Landscape unit reader: `lsu_read_elements.f90`; types `landscape_units` and `landscape_elements` in `calibration_data_module.f90`
- Landscape unit and basin output: `lsu_output.f90`, `lsu_carbon_output.f90`, `basin_output.f90`, `basin_sw_init.f90` (called from `time_control.f90`), file opening in `output_landscape_init.f90`
- Other uses of `ls_unit.ele` and `ls_unit.def`: `lcu_read_softcal.f90`, `hru_fr_change.f90` (action `hru_fr_update` in `actions.f90`), `gwflow_rech.f90`, `gwflow_gwet.f90`
- Aquifer elements: `aqu_read_elements.f90`, called from `proc_cal.f90`; basin aquifer output in `basin_aquifer_output.f90`, files opened in `header_write.f90`

File	Column	Fortran variable
`rout_unit.def`	`id`	not stored
`rout_unit.def`	`name`	`ru_def%name`
`rout_unit.def`	`elem_tot`	`nspu`
`rout_unit.def`	`elements`	`elem_cnt`, expanded into `ru_def%num` and `ru_def%num_tot`
`rout_unit.ele`	`id`	index of `ru_elem`
`rout_unit.ele`	`name`	`ru_elem%name`
`rout_unit.ele`	`obj_typ`	`ru_elem%obtyp`
`rout_unit.ele`	`obj_id`	`ru_elem%obtypno` (object number in `ru_elem%obj`)
`rout_unit.ele`	`frac`	`ru_elem%frac`
`rout_unit.ele`	`dlr`	`ru_elem%dr_name` (ratios in `ru_elem%dr`)
`rout_unit.rtu`	`id`	index of `ru`
`rout_unit.rtu`	`name`	`ru%name`
`rout_unit.rtu`	`define`	`ru%dbsc%elem_def`
`rout_unit.rtu`	`dlr`	`ru%dbsc%elem_dr`
`rout_unit.rtu`	`topo`	`ru%dbsc%toposub_db` (index `ru%dbs%toposub_db`)
`rout_unit.rtu`	`field`	`ru%dbsc%field_db` (index `ru%dbs%field_db`, values in `ru%field`)

File	Column	Fortran variable
ls_unit.def	line 2	mlsu, stored as db_mx%lsu_out
ls_unit.def	name	lsu_out%name
ls_unit.def	area	lsu_out%area_ha
ls_unit.def	elements	lsu_out%num, lsu_out%num_tot
ls_unit.ele	obj_typ	lsu_elem%obtyp
ls_unit.ele	obj_typ_no	lsu_elem%obtypno
ls_unit.ele	bsn_frac	lsu_elem%bsn_frac
ls_unit.ele	sub_frac	lsu_elem%ru_frac
aqu_catunit.ele	obj_typ, obj_typ_no	acu_elem%obtyp, acu_elem%obtypno
aqu_catunit.ele	bsn_frac, sub_frac, reg_frac	acu_elem%bsn_frac, acu_elem%ru_frac, acu_elem%reg_frac
aqu_catunit.ele	largest id	db_mx%aqu_elem

Purpose

The **link** line of file.cio names two optional files that connect objects outside the routing set up by the connectivity files:

- **chan-surf.lin** lists, for a SWAT+ channel, the HRUs or routing units on its floodplain. On days with overbank flow, the wetlands of those HRUs are filled from the channel.
- **aqu_cha.lin** lists, for an aquifer, the channels that receive its return flow. The flow is shared among the channels by channel length and drainage area.

The line gives the two names in this order. The defaults, **chan-surf.lin** and **aqu_cha.lin**, apply only when file.cio ends before the line (see the master file). Projects that do not use the files give **null** for both, as the reference datasets in the SWAT+ repository do. Earlier documentation called the first file **chan_surf.lin**.

Neither file has column names that SWAT+ checks. Both readers skip the header line and take the values by position, so the column names on this page are descriptive.

chan-surf.lin

Free format, whitespace-separated. Line 1 is a title and is skipped. Line 2 is a single whole number, the number of channel rows; SWAT+ also counts the rows itself, and the count only sets a minimum size for the table. Line 3 is a column header and is skipped. Each later line links one channel to its floodplain objects:

#	Column	Units	Description
1	id	-	Number of the SWAT+ channel, as numbered in chandeg.con.
2	name	-	Name of the floodplain. Not used in calculations.
3	num_obj	count	Number of floodplain objects that follow. A row with 0 gives the channel no floodplain links.
4	obj_typ	-	Type of the first floodplain object: **hru** or **ru** (routing unit).
5	obj_id	-	Number of the first floodplain object, as numbered in hru.con or rout_unit.con.
6, 7, ...	obj_typ, obj_id	-	One further pair for each remaining object, **num_obj** pairs in all.

Object types other than **hru** and **ru** are read and then ignored. For an **ru** object, SWAT+ takes the element numbers that rout_unit.def lists for that routing unit and uses each one as an HRU number. This selects the routing unit's HRUs only when each element number in rout_unit.ele equals the number of the HRU it refers to.

The links apply to channels routed with the SWAT+ channel method (chandeg.con). Legacy channels (channel.con) make no use of them.

Overbank flow to floodplain wetlands

Overbank flow happens on days when the channel's peak flow rate is above the bankfull threshold described under floodplain deposition. The overbank volume V_{ob} is the floodplain volume of the channel's rating curve at the peak flow rate. After floodplain deposition, SWAT+ goes through the floodplain HRUs in the order listed (a routing unit contributes its HRUs in the order of its elements). For each HRU it takes the room left in the HRU's wetland,

$$V_{fill} = V_{emr} - V_{wet}$$

where V_{emr} is the wetland's emergency volume and V_{wet} its current storage (m^3), and subtracts it from the overbank volume still available. When the overbank volume runs out, V_{fill} is cut to what was left. The wetland then receives the fraction

$$r = \min\left(1,\ \frac{V_{fill}}{Q_{in}}\right)$$

of the channel's inflow Q_{in} for the day (m^3), with the same fraction of its sediment, nutrients and other constituents, and the channel inflow is reduced by that fraction.

Only HRUs with surface storage, a `surf_stor` entry in `hru-data.hru` that names a record in `wetland.wet`, receive water. Other HRUs in the list take nothing from the channel. The water each wetland receives is reported in the `overbank` column (mm) of the HRU water balance output. See also overbank flow and floodplain deposition in the model theory, and the wetland inputs for the emergency volume.

Missing file and repeated read

SWAT+ reads `chan-surf.lin` while it sets up the SWAT+ channels, whether or not the model has any.

- If the slot is `null`, the file is skipped and no channel has floodplain links.
- If the slot gives a name and no file of that name exists, SWAT+ opens it anyway. This creates an empty file of that name in the run folder, nothing is read from it, and no channel has floodplain links.
- When `object.cnt` counts legacy channels (`cha` above 0), SWAT+ reads the file a second time, earlier in the setup, before the SWAT+ channels exist. If the file exists, the second read tries to set up the same table again and the run stops with an error.

aqu_cha.lin

Free format, whitespace-separated. Line 1 is a title and line 2 a column header; both are skipped. Each later line links one aquifer to a set of channels:

#	Column	Units	Description
1	`id`	-	Number of the aquifer, as numbered in `aquifer.con`.
2	`name`	-	Name of the link. Not used in calculations.
3	`num_elem`	count	Number of values in the channel list that follows. A row with 0 links no channels.
4, 5, …	`elements`	-	Channel numbers, as numbered in `chandeg.con`. A negative value marks a range: `1 -4` means channels 1 to 4.

The aquifer numbers must lie between 1 and the number of aquifers. SWAT+ reads as many rows as the largest aquifer number in the first column, so each aquifer should appear on one row only. Earlier documentation described the file as a replacement for `aquifer.con`. In 62.0.1 `aquifer.con` is still required, see overlap with aquifer outflows.

Return flow shares

For each linked aquifer, SWAT+ sorts the linked channels by drainage area, smallest first. The drainage area is the `area` column of `chandeg.con` and the channel length L_i is the `len` column of `hyd-sed-lte.cha` (km). With L_{tot} the total length of the linked channels, the length left when channel i and all smaller channels have stopped receiving flow is

$$L_{left,i} = L_{tot} - \sum_{j \le i} L_j$$

Each day the contributing channel length grows with the aquifer's return flow Q (mm):

$$L_c = L_{tot} \frac{Q}{\text{bf_max}}$$

where `bf_max` is given in `aquifer.aqu` (mm; see aquifer files). The first contributing channel k is the smallest channel with $L_c \ge L_{left,k}$, and each channel from k upward receives the share

$$f_i = \frac{L_i}{L_{left,k} + L_k}, \qquad i \ge k$$

while the smaller channels receive nothing. The shares of the contributing channels add up to 1, so the whole return flow of the aquifer is passed on. When Q is at or above `bf_max`, every linked channel receives flow in proportion to its length. As Q falls, the channels with the smallest drainage areas stop receiving flow first. See baseflow in the model theory.

Each linked channel adds its share of the aquifer's return flow hydrograph, with its nitrate, phosphorus, salts and constituents, to its inflow. This happens when the channel is routed: on the same day if the aquifer comes before the channel in the routing order, otherwise on the next day. The water appears in the `geo_bf` column (m^3/s) of the channel output.

Channel order, lengths and bf_max

- The shares are computed in drainage-area order. Each channel looks up its share by its position in the `aqu_cha.lin` list. A channel receives its intended share only when the list for each aquifer is already in order of increasing drainage area. In any other order, channels receive each other's shares; the total for the aquifer does not change.
- The channel length is taken from the row of `hyd-sed-lte.cha` whose position equals the channel's record number in `channel-lte.cha`. This gives the right length when both files list the channels in the same order.
- Once the file is read, the calculation divides by `bf_max` for every aquifer, including aquifers the file does not list. With `bf_max` = 0 the division fails, and builds that stop on floating-point errors (the Intel builds, for example) stop the run. Give every aquifer a `bf_max` above zero when `aqu_cha.lin` is used.

Overlap with aquifer outflows

`aqu_cha.lin` is read only when `object.cnt` counts aquifers, right after `aquifer.con`. The aquifer still routes its return flow through the outflows listed in `aquifer.con`, and SWAT+ also passes the shares above to the linked channels. A model that lists the same channels as outflows in `aquifer.con` and in `aqu_cha.lin` sends the return flow to them twice.

When gwflow is active, SWAT+ sets the aquifer count to 0 and `aqu_cha.lin` is not read.

Missing or short file

- If the slot is `null` or the file does not exist, the file is skipped and no aquifer is linked to channels.
- If the file exists but has fewer than two lines, the run stops with an end-of-file error.

Example

No dataset in the SWAT+ repository uses either file. The two files below are illustrative.

```
chan-surf.lin: illustrative example
2
id  name    num_obj  obj_typ  obj_id  obj_typ  obj_id
1   fp001   2        hru      4       hru      5
3   fp003   1        ru       2
```

Channel 1 is linked to HRUs 4 and 5, and channel 3 to the HRUs of routing unit 2. On a day with overbank flow in channel 1, the wetland of HRU 4 is filled first, then that of HRU 5, as long as overbank volume remains. An HRU without surface storage receives nothing.

```
aqu_cha.lin: illustrative example
id  name    num_elem  elements
1   aqu01   2         1  -4
2   aqu02   3         5  7  9
```

Aquifer 1 shares its return flow among channels 1 to 4, and aquifer 2 among channels 5, 7 and 9. For the shares to reach the intended channels, channels 1 to 4 must have increasing drainage areas in that order, and so must channels 5, 7 and 9. Neither aquifer should also route its return flow to these channels in `aquifer.con`.

Related files

- `file.cio`: names both files on the `link` line (master file).
- `chandeg.con`, `aquifer.con`, `hru.con`, `rout_unit.con`, `rout_unit.def` and `rout_unit.ele`: number the objects the files refer to (connectivity files).
- `channel-lte.cha` and `hyd-sed-lte.cha`: channel length and bankfull flow (channel tables).
- `aquifer.aqu`: `bf_max` and the return flow (aquifer files).
- `hru-data.hru`, `wetland.wet` and `hydrology.wet`: the surface storage that receives overbank water (HRU data, wetland inputs).
- `object.cnt`: the aquifer and legacy channel counts (object counts).

Sources

- File name slots: `type input_link` in `input_file_module.f90`, read in `readcio_read.f90`
- Readers: `overbank_read.f90` (`chan-surf.lin`), `aqu2d_read.f90` (`aqu_cha.lin`), with element ranges expanded in `define_unit_elements.f90`
- Call sites: `hyd_connect.f90` (both readers), `proc_cha.f90` (`overbank_read` again, `sd_channel_surf_link`, `aqu2d_init`); aquifer count reset for gwflow in `basin_read_objs.f90`
- Type definitions: `floodplain_parameters` in `sd_channel_module.f90`; `channel_aquifer_elements` and `geomorphic_baseflow_channel_data` in `hydrograph_module.f90`
- Floodplain HRU lists: `sd_channel_surf_link.f90`; routing unit elements from `ru_read_elements.f90`
- Overbank filling of wetlands: `sd_channel_sediment3.f90`; floodplain wetland storage in `ch_rtmusk.f90`; `overbank` output in `hru_control.f90`
- Channel sorting and lengths: `aqu2d_init.f90`
- Daily shares: `aqu_1d_control.f90`; added to channel inflow in `sd_channel_control3.f90`

File	Column	Fortran variable
`chan-surf.lin`	line 2 count	`mcha_sp`
`chan-surf.lin`	`id`	index `i` of `sd_ch`
`chan-surf.lin`	`name`	`sd_ch%fp%name`
`chan-surf.lin`	`num_obj`	`nspu`, `sd_ch%fp%obj_tot`
`chan-surf.lin`	`obj_typ`	`sd_ch%fp%obtyp`
`chan-surf.lin`	`obj_id`	`sd_ch%fp%obtypno`
`aqu_cha.lin`	`id`	index `iaq` of `aq_ch`
`aqu_cha.lin`	`name`	`aq_ch%name`
`aqu_cha.lin`	`num_elem`	`nspu`

File	Column	Fortran variable
aqu_cha.lin	elements	elem_cnt, expanded to aq_ch%num
aquifer.aqu	bf_max	aqudb%bf_max
hyd-sed-lte.cha	len	sd_chd%chl
chandeg.con	area	ob%area_ha

Purpose

This page covers three HRU parameter files and one channel file:

- hydrology.hyd: lateral flow, canopy storage, evaporation, percolation and enrichment parameters.
- topography.hyd: slope, slope lengths and the deposition coefficient. Also used by routing units.
- field.fld: field length and width for routing water and sediment across the landscape. Also used by routing units.
- shade_factor.shf: daily shade factors for the channel water temperature routine.

An HRU picks one record from each of the first three files by name, in the hydro, topo and field columns of hru-data.hru. hydrology.hyd, topography.hyd and field.fld are named on the hydrology line of file.cio. shade_factor.shf is not listed in file.cio; SWAT+ looks for a file with that name in the project folder.

All four files are free format and whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one record. The readers do not match columns by header name, so every value must be present and in the order given. Record names are stored with at most 16 characters.

An unmatched or misspelt hydro or topo name in hru-data.hru makes the HRU use the first record of the file without a message (see name lookup). hydrology.hyd and topography.hyd must therefore exist and must contain every name used.

hydrology.hyd

#	Column	Units	Typical range	Description
1	name	-	-	Name of the hydrology record, matched against the hydro column of hru-data.hru.
2	lat_ttime	days	0 to 180	Lateral flow travel time, called lat_time in earlier documentation. 0 lets SWAT+ compute it from the soil. See Lateral flow.
3	lat_sed	mg/L	0 to 5000	Sediment concentration in lateral and tile flow.
4	can_max	mm H_2O	0 to 100	Maximum canopy storage when the canopy is fully developed.
5	esco	-	0.01 to 1	Soil evaporation compensation factor. See Soil evaporation.
6	epco	-	0.01 to 1	Plant uptake compensation factor.
7	orgn_enrich	-	0 to 5	Organic N enrichment ratio for loading with sediment. 0 lets SWAT+ compute it for each runoff event.
8	orgp_enrich	-	0 to 5	Organic and sediment-attached P enrichment ratio for loading with sediment. 0 lets SWAT+ compute it for each runoff event.
9	cn3_swf	-	0 to 1	Soil water factor for the moisture condition III curve number: 0 puts it at field capacity, 1 at saturation.
10	bio_mix	-	0 to 1	Biological mixing efficiency.
11	perco	-	0 to 1	Percolation coefficient for the bottom soil layer. See Percolation from the soil profile.

#	Column	Units	Typical range	Description
12	`lat_orgn`	mg/L	0 to 200	Organic N concentration in lateral flow.
13	`lat_orgp`	mg/L	0 to 200	Organic P concentration in lateral flow.
14	`pet_co`	-	0.8 to 1.2	Multiplier on potential evapotranspiration. Older files call this column `harg_pet`. See Potential evapotranspiration.
15	`latq_co`	-	0 to 1	Lateral flow coefficient. See Lateral flow.

For an HRU with tile drains (set in its `landuse.lum` record), SWAT+ replaces `cn3_swf` with 0.95 and `perco` with 0.1 at the start of the run, whatever the record gives, and calibration changes to these two parameters are skipped for that HRU.

Lateral flow

Lateral flow is computed for each soil layer below the first with a hillslope storage method:

$$q_{lat,ly} = 0.024 \text{ latq_co } \frac{2\, SW_{ex,ly}}{\phi_{d,ly}/h_{ly}}\, K_{sat,ly}\, \frac{slp}{L_{lat}}$$

where $SW_{ex,ly}$ is the water above field capacity in the layer (mm), $\phi_{d,ly}$ the drainable porosity (saturation minus field capacity, mm), h_{ly} the layer thickness (mm), $K_{sat,ly}$ the saturated hydraulic conductivity (mm/h), slp the `slp` value from `topography.hyd` and L_{lat} its `lat_len` (m). The result is limited to the water above field capacity. The top layer produces no lateral flow. `latq_co` scales lateral flow linearly; 0 turns it off.

Lateral flow does not leave the HRU on the day it is generated. A fraction of the stored lateral flow is released each day:

$$f_{lat} = 1 - \exp\left(-\frac{1}{TT_{lat}}\right)$$

where TT_{lat} is `lat_ttime` in days. When `lat_ttime` is 0, SWAT+ computes it from the soil and the lateral slope length:

$$TT_{lat} = \max\left(1,\ 10.4\, \frac{L_{lat}}{K_{sat,mx}}\right)$$

with $K_{sat,mx}$ the highest saturated hydraulic conductivity in the profile (mm/h). Set `lat_ttime` to a specific value only when the base flow behaviour of the watershed is well known.

Soil evaporation

The evaporative demand on the soil is distributed with depth. The cumulative demand down to depth z (mm) is

$$E_z = E_s\, \frac{z}{z + \exp(2.374 - 0.00713\, z)}$$

where E_s is the soil evaporation demand for the day. A layer whose top is at depth z_u and bottom at z_l is asked for

$$E_{ly} = E_{z_l} - (1 - \text{esco})\, E_{z_u}$$

so with `esco` = 0 each layer meets only its share of the depth distribution, and as `esco` increases more of the demand can be met from lower layers. Layers below the soil evaporation depth do not take part. The demand on a layer is further reduced when the layer is drier than field capacity.

Plant uptake compensation

`epco` is copied to every plant of the HRU's plant community when the plants are initialized. As `epco` approaches 1, more of the water uptake demand can be met by lower layers when upper layers are dry; as it approaches 0, uptake stays closer to the original depth distribution.

Enrichment ratios

When `orgn_enrich` or `orgp_enrich` is above 0.001, it is used as a fixed enrichment ratio for organic N, or for organic and sediment-attached P, for every runoff event. Otherwise SWAT+ computes the enrichment ratio for each event from the sediment concentration in surface runoff. The enrichment ratio is the ratio of the concentration of the constituent in the transported sediment to its concentration in the soil surface layer; it is above 1 because the sediment in runoff contains a larger share of the fine particles that the nutrients are attached to.

Curve number soil water factor

`cn3_swf` sets the soil water content at which the curve number for moisture condition III applies:

$$SW_{CN3} = FC + \text{cn3_swf}\,(SAT - FC)$$

where FC and SAT are the profile water contents at field capacity and saturation. Larger values delay the onset of surface runoff after dry periods. See hydrology in the model theory.

Biological mixing

Biological mixing redistributes residue and nutrients in the soil through the activity of earthworms and other soil biota, using the same method as a tillage operation with `bio_mix` as the mixing efficiency. With `carbon` = 0 in `codes.bsn`, it is applied once at the end of every calendar year, down to 50 mm or the bottom of the profile if that is shallower. With the other `carbon` codes this column is not used; `carbon` = 2 applies a daily biological mixing with an efficiency taken from the tillage data.

Percolation from the soil profile

`perco` limits percolation out of the bottom soil layer, for example to represent an impermeable layer or a high water table. Percolation between the upper layers is not affected. The potential percolation from the bottom layer is multiplied by

$$f_{perc} = \min[1,\ \exp(-\,(1.0052\ \ln(-\ln \text{perco}) + 5.6862))]$$

The relation is strongly non-linear:

`perco`	0	0.1	0.5	0.8	0.9	0.95	0.99	0.999 and above
fraction of potential percolation	0	0.002	0.005	0.015	0.033	0.067	0.35	1

So 0 stops percolation from the profile, values up to about 0.9 allow only a few percent of it, and the restriction is lifted only as `perco` approaches 1.

Potential evapotranspiration

`pet_co` multiplies the potential evapotranspiration of the HRU, whichever method is selected in `codes.bsn` (Priestley-Taylor, Penman-Monteith, Hargreaves or values read from file). A value of 0 is treated as 1. Values below 0.01 are taken as the old Hargreaves coefficient written to `harg_pet` by earlier editors and divided by 0.0023, so 0.0023 becomes 1.0.

Canopy storage

The canopy storage available on a day is

$$CAN_{day} = \text{can_max}\,\frac{LAI}{LAI_{mx}}$$

where LAI is the current leaf area index of the plant community and LAI_{mx} its maximum. Precipitation fills the canopy before it reaches the ground.

Sediment and organic nutrients in lateral flow

`lat_sed` adds sediment to the HRU sediment yield in proportion to lateral and tile flow, and `lat_orgn` and `lat_orgp` add organic N and P in proportion to lateral flow. In 62.0.1 the sediment from `lat_sed` is added twice: once as a mass computed from the HRU area and once again as a value that does not use the area. Sediment yields from HRUs with a non-zero `lat_sed` therefore include an extra term. The conversion used for `lat_orgn` and `lat_orgp` is 100 times smaller than mg/L of lateral flow would give, so the loads are 1% of what the stated units suggest.

topography.hyd

#	Column	Units	Built-in value	Description
1	`name`	-	`default`	Name of the topography record, matched against the `topo` column of `hru-data.hru` and of `rout_unit.rtu`.
2	`slp`	m/m	0.02	Average slope steepness, usually computed from the DEM. Used for lateral flow, the USLE slope-length factor, time of concentration and sediment transport across the HRU.
3	`slp_len`	m	50	Average slope length for erosion: the distance over which sheet flow is the main surface runoff process, measured to the point where flow begins to concentrate. Used in the USLE slope-length factor. The full value is used: 62.0.1 does not apply the maximum slope length of the HRU's conservation practice.
4	`lat_len`	m	50	Slope length for lateral subsurface flow, usually longer than `slp_len`. A value of 0 is replaced by 50 m.
5	`dist_cha`	m	100	Average distance to the stream. Not used.

#	Column	Units	Built-in value	Description
6	`depos`	-	1	Deposition coefficient for sediment transport capacity when surface runoff from another object is routed across the HRU. A larger value gives a higher transport capacity, so more sediment leaves the HRU.

The built-in values apply only to a record that is never read; an HRU always uses a record from the file.

When runoff from upslope objects is routed across an HRU, its sediment transport capacity is proportional to

$$\text{depos}\ C\ slp^{1.4}\ w^{1.4}$$

and to the overland flow, where C is the USLE cover factor of the HRU and w the field width `wd` from `field.fld`.

field.fld

#	Column	Units	Built-in value	Description
1	`name`	-	`default`	Name of the field record, matched against the `field` column of `hru-data.hru` and of `rout_unit.rtu`.
2	`len`	m	500	Field length. Used for the time of concentration of routing units; not used for HRUs.
3	`wd`	m	100	Field width. Used in the sediment transport capacity when surface runoff is routed across an HRU.
4	`ang`	degrees	30	Field angle. Not used.

An HRU whose `field` entry is `null`, or a name that is not found, uses the built-in values.

shade_factor.shf

#	Column	Units	Description
1	`jday`	day of year	Day the value applies to (1 to 365).
2	`cha`	-	Channel number (row in `chandeg.con`). In revisions before 62.0.1 this column was a landscape unit number.
3	`value`	fraction	Shade factor, the fraction of incoming solar radiation blocked before it reaches the water surface.

The file has no name column and is read as a list of (day, channel, value) rows. The channel water temperature routine reduces short-wave radiation at the water surface by the factor (1 − value). On day 366 of leap years the factor is 0.5.

The file is meant to be used when `sf_on` = 1 in `temperature.cha`. In 62.0.1 the temperature routine takes all its settings from the built-in record instead of from the rows of `temperature.cha`. The built-in record has `sf_on` = 0 and `ssff` = 0.5, so the shade factor is 0.5 for every channel and day, and `shade_factor.shf` is read but not used. No example dataset in `refdata/` has a `shade_factor.shf` file.

Calibration

These parameters can be changed during calibration through `calibration.cal` (see parameter change files), using these names: `lat_ttime`, `lat_sed`, `canmx` (`can_max`), `esco`, `epco`, `erorgn` (`orgn_enrich`), `erorgp` (`orgp_enrich`), `cn3_swf`, `biomix` (`bio_mix`), `perco`, `lat_orgn`, `lat_orgp`, `petco` (`pet_co`), `latq_co`, `slope` (`slp`), `slope_len` (`slp_len`), `lat_len`, `dis_stream` (`dist_cha`), `field_len` (`len`), `field_wid` (`wd`) and `field_ang` (`ang`).

A change to `lat_ttime` is applied to the daily release fraction f_{lat}, which SWAT+ has already computed from the input, and the result is then converted again as if it were a travel time in days. The effect of a `lat_ttime` change is therefore not the effect of changing the travel time in the file by the same amount. Changes to `dis_stream` and `field_ang` have no effect, because the values are not used.

Examples

`refdata/Ames_sub1/hydrology.hyd` in the SWAT+ repository (first two rows):

```
hydrology.hyd:
name                lat_ttime       lat_sed       can_max          esco          epco   orgn_enrich   org
hyd0001               0.00000       0.00000       0.00000       0.95000       1.00000       0.00000
hyd0002               0.00000       0.00000       0.00000       0.95000       1.00000       0.00000
```

Lateral flow travel time and enrichment ratios are computed by SWAT+, there is no canopy storage, and `cn3_swf` = 0 puts the moisture condition III curve number at field capacity. `perco` = 0.5 lets only about 0.5% of the potential percolation leave the bottom of the profile (unless the HRU is tile drained, which resets `perco` to 0.1), and `latq_co` = 0.01 keeps lateral flow small.

`refdata/Osu_1hru/hydrology.hyd` uses the older column header `harg_pet` with the value 0, which SWAT+ treats as a `pet_co` of 1.

`refdata/Ames_sub1/topography.hyd` (first two rows):

```
topography.hyd:
name                        slp       slp_len       lat_len      dist_cha         depos
topohru0001             0.02500      91.46300      91.46300     100.00000       1.00000
topohru0002             0.02500      91.46300      91.46300     100.00000       1.00000
```

`refdata/Osu_1hru/field.fld` (first two rows):

```
field.fld: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                        len            wd           ang
fld011                500.00000     100.00000      30.00000
fld012                500.00000     100.00000      30.00000
```

Related files

- `hru-data.hru`: selects the records by name in its `hydro`, `topo` and `field` columns.
- `file.cio`: names the three HRU files on its `hydrology` line.
- `landuse.lum`: sets tile drainage, which overrides `cn3_swf` and `perco`.
- `codes.bsn`: PET method and `carbon` code that `pet_co` and `bio_mix` depend on.
- `temperature.cha`: channel temperature settings, including `sf_on` for `shade_factor.shf`.
- Hydrology and soil water in the model theory.

Sources

- Readers: `hydrol_read.f90`, `topo_read.f90`, `field_read.f90`, `shade_factor_read.f90` (called from `proc_read.f90`)
- Type definitions: `type hydrology_db` in `hydrology_data_module.f90`; `type topography_db` and `type fields_db` in `topography_data_module.f90`; `type shade_factor_data` in `hydrograph_module.f90`
- File names: `type input_hydrology` and `type shade_factor` in `input_file_module.f90`, `in_hyd` read in `readcio_read.f90`

- Copy to each HRU, tile override, `pet_co` and `perco` conversion: topohyd_init.f90; routing units in ru_read.f90
- Lateral flow: hydro_init.f90, swr_percmicro.f90, swr_substor.f90
- Evaporation, PET and uptake: et_act.f90, et_pot.f90, plant_init.f90
- Canopy: sq_canopyint.f90; curve number: curno.f90
- Enrichment ratios: nut_orgn.f90, nut_orgnc2.f90, nut_psed.f90
- Lateral sediment and organic nutrients: swr_latsed.f90
- Biological mixing: time_control.f90, mgt_newtillmix_cswat0.f90
- Landscape routing and time of concentration: rls_routesurf.f90, time_conc_init.f90
- Shade factor use: ch_temp.f90, ch_read_temp.f90, `type channel_temperature_data` in channel_data_module.f90
- Calibration: cal_parm_select.f90

File	Column	Fortran variable
hydrology.hyd	`lat_ttime`	`hyd_db%lat_ttime`
hydrology.hyd	`lat_sed`	`hyd_db%lat_sed`
hydrology.hyd	`can_max`	`hyd_db%canmx`
hydrology.hyd	`esco`	`hyd_db%esco`
hydrology.hyd	`epco`	`hyd_db%epco`
hydrology.hyd	`orgn_enrich`	`hyd_db%erorgn`
hydrology.hyd	`orgp_enrich`	`hyd_db%erorgp`
hydrology.hyd	`cn3_swf`	`hyd_db%cn3_swf`
hydrology.hyd	`bio_mix`	`hyd_db%biomix`
hydrology.hyd	`perco`	`hyd_db%perco` (converted to `hru%hyd%perco_lim`)
hydrology.hyd	`lat_orgn`	`hyd_db%lat_orgn`
hydrology.hyd	`lat_orgp`	`hyd_db%lat_orgp`
hydrology.hyd	`pet_co`	`hyd_db%pet_co`
hydrology.hyd	`latq_co`	`hyd_db%latq_co`
topography.hyd	`slp`	`topo_db%slope`
topography.hyd	`slp_len`	`topo_db%slope_len`
topography.hyd	`lat_len`	`topo_db%lat_len`
topography.hyd	`dist_cha`	`topo_db%dis_stream`
topography.hyd	`depos`	`topo_db%dep_co`
field.fld	`len`	`field_db%length`
field.fld	`wd`	`field_db%wid`
field.fld	`ang`	`field_db%ang`
shade_factor.shf	`jday`, `cha`, `value`	`shf_db%jday`, `shf_db%cha`, `shf_db%value`

Purpose

A recall object is a spatial object whose outflow is read from a file instead of simulated. It is used for point sources such as wastewater treatment plants, for inflow from outside the modelled area (an upstream gauge or another model), and for diversions. Each day the recall object sends the flow and loads for that day to the objects named in its row of recall.con, in the same way an HRU or channel sends its outflow downstream.

A recall object needs three pieces of input:

1. A row in recall.con, which gives its location and the receiving object (usually a channel, `sdc`).
2. A record in `recall.rec` with the same `id`, which gives the time step and the name of the data file.
3. The data file with the flows and loads.

The number of recall objects is the `rec` column of object.cnt. `recall.rec` is named on the `recall` line of file.cio (default `recall.rec`; `null` turns recall input off).

recall.rec

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one recall record with four values. The header names are not checked; the example in `refdata/Osu_1hru`

uses `NUM NAME TYPE FNAME`. Anything after the fourth value on a line is ignored.

#	Column	Units	Description
1	`id`	-	Record number. A recall object uses the record whose `id` equals its `id` in `recall.con`; the `rec` column of `recall.con` is not used. The largest `id` sets the size of the table.
2	`name`	-	Name of the record, up to 25 characters.
3	`typ`	-	Time step of the data: 0 = sub-daily, 1 = daily, 2 = monthly, 3 = yearly, 4 = constant taken from an export coefficient record.
4	`filename`	-	Name of the data file (up to 25 characters), or for `typ` 4 the name of a record in `exco.exc`.

In revision 8a1edd4, recall objects were read from `recall_db.rec`, with six data file names (organic and mineral, pesticide, pathogen, heavy metal, salt and constituent) each followed by its units and a time step word (`sub`, `day`, `mo`, `yr`), and `recall.rec` was not read. In 62.0.1 recall objects are read from `recall.rec` with one data file and an integer time step per record. `recall_db.rec` is still read, with plain file names, but only for the uses listed under recall_db.rec.

Recall data file

One data file per recall record, named in `recall.rec`.

- Line 1: title, skipped.
- Line 2: number of years in the file. Read but not used; SWAT+ works out the years from the rows.
- Line 3: column header, skipped.
- Each later line: one time step.

Each data line has six leading values followed by 18 values in this order:

#	Column	Units	Description
1	`jday`	-	Day of year (1 to 366). Selects the day for daily data.
2	`mo`	-	Month (1 to 12). Selects the month for monthly data.
3	`day_mo`	-	Day of month. Not used.
4	`yr`	-	Year.
5	`ob_typ`	-	Object type label. Not used.
6	`ob_name`	-	Object name label. Not used.
7	`flo`	m^3	Water volume for one day.
8	`sed`	t	Sediment.
9	`orgn`	kg N	Organic N.
10	`sedp`	kg P	Organic P.
11	`no3`	kg N	Nitrate N.
12	`solp`	kg P	Soluble (mineral) P.
13	`chla`	kg	Chlorophyll-a.
14	`nh3`	kg N	Ammonium N.
15	`no2`	kg N	Nitrite N.
16	`cbod`	kg	Carbonaceous biological oxygen demand.
17	`dox`	kg	Dissolved oxygen.
18	`san`	t	Detached sand.
19	`sil`	t	Detached silt.
20	`cla`	t	Detached clay.
21	`sag`	t	Detached small aggregates.
22	`lag`	t	Detached large aggregates.
23	`grv`	t	Gravel.
24	`temp`	°C	Water temperature.

The values are used as given, as the recall object's outflow for one day. For daily data this is the day's volume and loads. Monthly and yearly values are also applied on every day of their month or year, so give them as daily

amounts: the mean daily volume and load for the month or year. A monthly or yearly total would be applied once for every day of the period. Pesticides, pathogens, heavy metals, salts and other constituents cannot be given in this file; their recall loads are zero.

Time steps

`typ`	Rows	Value used on a simulation day
1, daily	One row per day	The row with that day of year (`jday`) in that year.
2, monthly	One row per month	The row with that month (`mo`) in that year.
3, yearly	One row per year	The row for that year.
0, sub-daily		Does not work, see below.
4, constant	No data file	Does not work, see below.

Years

SWAT+ reads the first data row to find the first year in the file.

- If the file starts in or before the first simulation year, the rows before that year are skipped. The file must then contain the first simulation year: if it ends before that year, the reader keeps searching and the run does not get past reading the inputs.
- If the file starts after the first simulation year, its first year is placed at the matching simulation year and earlier years get no recall input.

From there each change of year in the file moves to the next simulation year, and reading stops at the last simulation year. A year missing from the middle of the file is not skipped over: the following years shift one year earlier. Give every year from the start of the data to the end without gaps.

For daily and monthly data, the recall object sends nothing on days outside the years covered by the file. For yearly data the test of the year range always passes, but years without data have zero values, so the effect is the same.

Diversions

A negative `flo` in daily data removes water. When the recall object's first connection in `recall.con` is a channel with more than 10 m^3 of water, SWAT+ also removes nitrate, soluble P, ammonium, nitrite, dissolved oxygen and organic N from that channel at the channel's concentration, limited to what is in the channel. Salt and constituent mass are handled in the same way when those modules are active.

Sub-daily and constant records

Sub-daily records (`typ` 0) do not work in 62.0.1. The reader has no step that reads a sub-daily data line, so it keeps returning to the same line and never reaches the end of the file; a run with a sub-daily recall record does not get past reading the inputs.

Constant records (`typ` 4) are meant to take a fixed daily hydrograph from the `exco_om.exc` row of the `exco.exc` record named in `filename`. In 62.0.1 `recall.rec` is read before the export coefficient files, so the name is looked up in an empty list and the constant hydrograph stays zero. The daily recall step also has no branch for `typ` 4. A recall object with a `typ` 4 record therefore sends nothing. Use an export coefficient object in `exco.con` for a constant source instead.

recall_db.rec

`recall_db.rec` is a second recall list, read when a file with this name is in the project folder. It is not named in `file.cio`. It is used by the water allocation files and by reservoir release decision tables, not by recall objects in the routing.

Line 1 is a title and line 2 the header; both are skipped. Each later line has nine values:

#	Column	Description
1	id	Record number.
2	name	Record name, up to 13 characters.
3	org_min	Organic and mineral data file name.
4	pest	Pesticide data file name.
5	path	Pathogen data file name.
6	hmet	Heavy metal data file name.
7	salt	Salt data file name.
8	constit	Other constituent data file name.
9	descrip	Description, one word or quoted.

All nine values are required; a missing value makes the reader continue on the next line. The file names are stored as text only. None of these files is opened through `recall_db.rec`.

The two uses in 62.0.1:

- A reservoir release decision table action of type `meas` names a `recall_db.rec` record. The reservoir then releases the daily, monthly or yearly flow of the `recall.rec` record with the same number, without the year-range test used for recall objects. Keep the numbering of the two files the same.
- Water use, water treatment and outside-source entries in the water allocation files can take their water quality from a recall record by matching their name against `org_min`. The link from a `recall_db.rec` record to the recall data is never set in 62.0.1, so this option does not deliver the intended data.

pest.com

If a file named `pest.com` is in the project folder, SWAT+ reads it as a list of pesticide recall files. The values it reads are stored in the flow and nutrient records of the recall objects with the same numbers, replacing the data read through `recall.rec`, and no pesticide load is set. Do not put a file with this name in the project folder.

Salt and constituent recall files

`salt_recall.rec` and `cs_recall.rec` give salt-ion and constituent loads for recall objects when the salt or constituent modules are active. Their readers are in the source, but 62.0.1 does not call them, so these files are not read. The daily recall step still looks up salt and constituent recall data whenever salts or constituents are simulated. A model that simulates salts or constituents and has recall objects is therefore likely to stop with a memory error. See salts and constituents.

For reference, the format the readers expect: a title, a header, then one row per recall object with `id`, `name`, `typ` (1 = daily, 2 = monthly, 3 = yearly, 4 = export coefficient) and `filename`. Each data file has a title (the word `Incoming` marks a source from outside the watershed), the number of years, a header, and rows of `jday`, `mo`, `day_mo`, `yr`, `ob_typ`, `ob_name` followed by one value per salt ion or constituent in the order of `constituents.cs`.

Region files

`rec_catunit.def`, `rec_reg.def` and `rec_catunit.ele` group recall objects into regions. They are named on the `regions` line of `file.cio` and read when the calibration inputs are set up.

- `rec_catunit.def` and `rec_reg.def`: a title, the number of regions, a header, then per region `id`, `name`, `area` (ha), the number of elements, and the element numbers. With 0 elements a region contains every recall object.
- `rec_catunit.ele`: a title, a header, then per element `id`, `name`, `obj_typ`, `obj_typ_no`, `bsn_frac`, `sub_frac` and `reg_frac`.

In 62.0.1 these regions are not used for any output or calibration step, and the tables they fill are not set up before reading, so a definition file with one or more regions is likely to stop the run. Set these entries to `null` in `file.cio`.

Output

Recall objects are printed to `recall_day.txt`, `recall_mon.txt`, `recall_yr.txt` and `recall_aa.txt` (and the matching `.csv` files), selected by the `recall` row of `print.prt`. The sum over all recall objects is printed to the `basin_psc` files. The columns are described in recall outputs.

Columns in earlier documentation

Earlier documentation listed `recall.rec` with the columns `id`, `name`, `om_name`, `om_unit`, `om_tstep`, `pest_name`, `pest_unit`, `pest_tstep`, `path_name`, `path_unit`, `path_tstep`, `hmet_name`, `hmet_unit`, `hmet_tstep`, `salt_name`, `salt_unit`, `salt_tstep`, `cst_name`, `cst_unit` and `cst_tstep`. That is the layout recall objects used before 62.0.1. SWAT+ 62.0.1 no longer reads these columns: `recall.rec` has the four columns described above, and `recall_db.rec` has plain file names without units or time steps.

Example

`refdata/Osu_1hru/recall.rec` in the SWAT+ repository:

```
recall.rec
NUM      NAME     TYPE      FNAME     OBJ_TYP      OBJ_NAME        flo      sed     orgn     sedp      no3     solp
   1     PSTYR     3       RECYR.         hru          hru1    12120.5        0        0        0        0        0
   1       1           1     2011         hru          hru2     9000.7        0        0        0        0        0
```

SWAT+ reads the four values `1 PSTYR 3 RECYR.`: record 1, named `PSTYR`, yearly data in a file named `RECYR.`. The rest of the line, and the second data line, are ignored, because the largest `id` is 1 and only one record is read. This dataset has `rec` = 0 in `object.cnt`, so the file is not used in the run, and `RECYR.` is not included.

A daily data file for one recall record looks like this (illustrative values, columns 8 to 24 shortened):

```
daily point source, wastewater plant 1
2
jday  mo  day_mo    yr  ob_typ  ob_name      flo     sed    orgn    sedp     no3    solp  ...  temp
   1   1       1  2010  sdc     cha05     4320.0   0.050    1.20    0.30    8.50    1.10  ...  12.0
   2   1       2  2010  sdc     cha05     4410.0   0.050    1.25    0.31    8.60    1.12  ...  12.0
```

Each row gives the volume (m^3) and loads (t, kg) discharged on that day, here about 0.05 m^3/s.

Related files

- `recall.con`: location of each recall object and the objects that receive its outflow.
- `object.cnt`: number of recall objects (`rec`).
- `file.cio`: names `recall.rec` on the `recall` line and the region files on the `regions` line.
- `print.prt`: recall output switches.
- Recall outputs: columns of the recall output files.
- Decision tables: reservoir release action `meas`.
- Water allocation: water quality from recall records.
- Recall, export coefficient and delivery ratio objects in the model theory.

Sources

- Reader for `recall.rec`, the data files and `pest.com`: `recall_read.f90`; reader for `recall_db.rec`: `recalldb_read.f90`. Both are called from `main.f90.in`, after `hyd_connect.f90` and before `exco_db_read.f90`
- Type definitions: `type recall_hydrograph_inputs` and `type hyd_output` in `hydrograph_module.f90`; `type recall_databases` in `recall_module.f90`
- Daily use: `case ("recall")` in `command.f90`; diversions in `recall_nut.f90`, `recall_salt.f90`, `recall_cs.f90`
- Salt and constituent readers (not called): `recall_read_salt.f90`, `recall_read_cs.f90`

- Region files: rec_read_elements.f90, called from proc_cal.f90; arrays in calibration_data_module.f90
- Uses of recall_db.rec: dtbl_res_read.f90, res_hydro.f90, water_use_read.f90, water_treatment_read.f90, water_osrc_read.f90, wallo_use.f90, wallo_treatment.f90, wallo_start_day.f90
- Export coefficient lookup: exco_read_om.f90
- File names: in_rec%recall_rec and in_regs%def_psc, def_psc_reg, ele_psc in input_file_module.f90; output files opened in header_write.f90

File	Column	Fortran variable
recall.rec	id	array index i of recall
recall.rec	name	recall(i)%name
recall.rec	typ	recall(i)%typ
recall.rec	filename	recall(i)%filename
data file	flo to temp	recall(i)%hd(istep, iyrs) (type hyd_output)
recall_db.rec	name, org_min, pest, path, hmet, salt, constit, descrip	recall_db(i)%name, %org_min, %pest, %path, %hmet, %salt, %constit, %descrip